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Exchange Rate Density Forecasting Using Quantile Regressions

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Abstract

We construct one-month-ahead conditional density forecasts for the USD/EUR exchange rate returns using the two-step quantile regression approach introduced by Adrian et al. (2019b). To construct conditional variables with predictive power for future exchange rate returns, we propose three indices based on a Principal Component Analysis (PCA), and four indices based on a Nelson-Siegel yield curve estimation. We show that the distribution of exchange rate returns evolves over time. By means of a Amisano & Giacomini (2007)-test, we show that applying the proposed technique out-of-sample, when separately using 6 out of 7 conditional variables, leads to a significant outperformance of a random walk density forecast at a 96-month rolling window.

1 Introduction

On average, more than 7.5 trillion US dollar is traded each day on the foreign exchange market, with over 1.7 trillion US dollar between the US dollar and the Euro (Bank for International Settlements, 2022). This research aims to improve the accuracy of one-month-ahead density forecasts of the USD/EUR exchange rate return[1], relative to a random walk model. With the exchange rate influencing the value of international revenues, and the price of exports and imports in home currency, the exchange rate is highly relevant, including for business, government, and investors such as pension funds.

Academic literature initially exhibited a strong focus on forecasting the level of exchange rates. However, an economic agent might be more interested in the tail-risk or entire distribution of future exchange rates. Additionally, as these 'prices of currencies' are highly volatile and contain much noise, providing an accurate point forecast is a difficult task. Meese & Rogoff (1983) indeed show that various exchange rate models based on fundamental macroeconomic concepts do not manage to outperform a random walk model in forecasting the level of exchange rates. Two decades later, Neely & Sarno (2002) showed that although more sophisticated econometric techniques for point forecasts were developed, the findings of Meese-Rogoff remained largely robust. The findings until then shed light on the underlying complexity of exchange rates and the weaknesses of economic and financial theory regarding exchange rate modeling.

Around the early 2000s, density forecasts gained prominence when Sarno & Valente (2005) and Hong et al. (2007) showed that sophisticated econometric models are able to outperform random walk models in forecasting the density of exchange rates. This result raised a new perspective in the so-called 'Meese-Rogoff puzzle', a frequently debated proposition that states that no exchange rate model can outperform a random walk model in out-of-sample forecasting. Besides, it uniquely focused on predicting the density of an exchange rate, rather than the level. A density forecast is more informative, and is thus an improvement for any economic agent that is interested in the future exchange rate.

Past academic literature has mainly focused on using complex time series models to improve density forecasts for exchange rates relative to a random walk model. Instead, our research

[1] In this paper, USD/EUR denotes the price of one US dollar expressed in euros (e.g., 0.91 €/$).

paper investigates the usage of a simple 2-step procedure introduced by Adrian et al. (2019b), by exploring the following research question:

Can the random walk model be beaten by a quantile regression forecasting approach in one-month-ahead out-of-sample density forecasts for the USD/EUR exchange rate?

The strength of the technique introduced by Adrian et al. (2019b) is that, if one finds an informative variable to condition on, a possibly accurate out-of-sample density forecast can be provided whilst retaining analytical simplicity. As a direct consequence, an essential ingredient for answering this research question is creating a conditional time series variable that contains some predictive power for the future USD/EUR exchange rates. For this conditional variable, we consider: (1) the first principal component [2] of various macroeconomic variables; (2) the first principal component of the one month log return of a major US stock price index, a major Eurozone stock price index, and of the oil price; (3) the first principal component of both these macroeconomic and stock and oil price variables; (4) the Nelson-Siegel relative level factor like Y.-C. Chen & Tsang (2013) between spot rates of AAA-rated Eurozone and US Treasury central government debt yields; (5) this relative level factor between spot rates of all Eurozone and US Treasury central government debt yields; (6) the Nelson-Siegel relative level slope like Y.-C. Chen & Tsang (2013) between spot rates of AAA-rated Eurozone adn US Treasury central government debt yields; (7) this relative slope factor between spot rates of all Eurozone central and US Treasury government debt yields;

We use the quantile regression framework of Adrian et al. (2019b) to generate one-month-ahead density forecasts for the USD/EUR exchange rate returns. We show that this technique can successfully be applied in out-of-sample forecasting when using our proposed indices as conditional variables. At the 8-year rolling window size, applying the quantile regression framework separately using each of the conditional indices, with the exception of the index incorporating stock price and oil data, yields a significant outperformance relative to a random walk density forecast. At the 6-year rolling window size, using the relative level factor of the Nelson-Siegel curve leads to a significant outperformance of the random walk model. As a result, our paper highlights the effectiveness of using macroeconomic and financial variables in forecasting future exchange rates.

Related Literature

Since the work of Meese & Rogoff (1983), it has been widely recognized that exchange rates are difficult to predict using traditional econometric models. Their findings, now referred to as the 'Meese-Rogoff puzzle', demonstrated that a simple random walk model often outperforms structural macroeconomic models in exchange rate forecasting. This result challenges the ability of structural economic models to provide useful out-of-sample forecasts, leading to further research

[2] This principal component refers to the standardized value of the first component obtained from a Principal Component Analysis (PCA) conducted on the full sample of conditional variables. Note that the PCA includes all available data—also from periods beyond the forecast horizon—which implies that the axes of the principal components are determined by 'future' data. Creating the index on earlier data (e.g., 1980–2000) introduces the concern that the economic environment may have changed substantially, limiting its relevance. Consequently, this presents a practical limitation of our research. The given definition in this footnote applies to all instances of 'the first principal component' in (1)–(3).

into alternative methods.

As noted by Rossi (2013), the majority of the literature regarding exchange rate forecasting has historically focused on point forecasts. However, substantial shortcomings of point forecasts have increasingly been recognized, particularly their inability to adequately capture tail risks associated with extreme exchange rate movements. Cenedese et al. (2014), for instance, demonstrate that extreme depreciations and appreciations exhibit fundamentally different dynamics compared to typical fluctuations, suggesting that standard mean forecasts systematically underestimate tail risks. Similarly, Crump et al. (2018) argue more broadly that modeling the full probability distribution of financial variables provides a better and more complete assessment of risk compared to level forecasts alone.

A new line of research that emerged in the early 2000s explicitly focused on density forecasts. One of the earliest studies by Diebold et al. (1999) used the RiskMetrics model by J.P. Morgan (1996) to calculate density projections for exchange rate returns. Furthermore, Sarno & Valente (2005) notably demonstrated that spot exchange rate density forecasts derived from a Markov-switching Vector Error Correction Model (VECM) were able to outperform a random walk density forecast. Expanding on these advancements, Hong et al. (2007) showed that exchange rate return density forecasts from univariate time series models explicitly accounting for fat-tailed distributions, time-varying volatility, and regime-switching behavior could successfully outperform the random walk model. Most of the models used in these earlier studies have considerable limitations in that they focus on short forecasting horizons, complex model structures and reliance on parametric estimation procedures.

In response to the complexity and restrictive assumptions inherent to traditional density forecasting models, quantile regression methods have gained popularity due to their flexibility, analytical simplicity, and reduced reliance on strict parametric structures. Gaglianone & Lima (2012) provided a clear methodological application of quantile regressions, demonstrating how predictive densities can be constructed from multiple quantile estimates through semi-parametric interpolation. Although their approach offers flexibility, it relies solely on interpolation and does not explicitly model distributional tails or impose a parametric structure, limiting interpretability and precision in capturing extreme risks.

Addressing these limitations, Adrian et al. (2019b) introduced a two-step approach for constructing conditional density forecasts. In this approach, conditional quantiles are estimated and fitted to a skewed-t distribution to construct predictive densities. This parametric fitting approach provides an analytically simpler alternative to previous density forecasting techniques such as those of Sarno & Valente (2005) and Hong et al. (2007). Besides, it does not rely on underlying parametric assumptions that previous techniques required a forecaster to make. Furthermore, a considerable benefit is that this technique allows us to make a density forecast of a dependent variable conditional on a set of independent variables, as opposed to previous unconditional density forecasting techniques. This uniquely enables the relevance of predictive indices. Extending the approach of Adrian et al., Yapi (2020) adapted the methodology specifically for exchange rate forecasting.

The quest for finding informative predictive indices for future exchange rates, an essential ingredient in techniques like that of Adrian et al. (2019b), essentially comes down to the long-debated puzzle: What drives exchange rates? Rossi (2013) notes that a clear consensus in the academic literature on this topic is absent. An example of a successful attempt is that of Eguren-Martin & Sokol (2022), who use a Dynamic Principal Component Analysis (PCA) - method to extract a global financial factor from a wide range of macro-economic and financial indicators. In an agnostic manner, this study combines a variety of variables that possibly drive exchange rates. This global financial factor is, when used as a conditional variable in the technique of Adrian et al. (2019b), indeed shown to contain predictive power for future exchange rate returns.

An alternative perspective on exchange rate predictability is provided by Y.-C. Chen & Tsang (2013). Following Nelson & Siegel (1987), they argue that the level, slope and curvature of the yield curve contains valuable information on future exchange rate movements. However, to our best knowledge, such yield-curve fundamentals have not yet been used to create conditional density forecasts.

The rest of this paper will be structured as follows. In Section 2, we provide an overview of the data that is used to conduct our analysis. In Section 3, we present the methodology of the quantile regression forecasting framework and testing procedures. In Section 4, we perform an in-sample analysis on our predictive indices and discuss univariate quantile regression results. In Section 5, we present the out-of-sample forecasting results and examine different risk metrics. We conclude and discuss limitations of our research in Section 6. The Appendix contains details on our data and methodology.

2 Data

We aim to examine whether one-month-ahead density forecasts for the USD/EUR exchange rate, based on the quantile regression framework, outperform those of a random walk model. To perform analysis and create conditional variables, we use several datasets.[3] First, we use the nominal exchange rate, for which we compute the log differences. Second, we use a set of variables, which we refer to as 'fundamental' and 'experimental', from July 2000 to April 2018 for the US and the Eurozone[4], aimed at capturing signals about the exchange rate returns and associated risks. Finally, we use yield curve data from September 2004 to December 2018 for US Treasuries, AAA-rated Eurozone central government bonds, and all Eurozone central government bonds. In the rest of this section, we provide details for the data we downloaded and transformed. Table 1 provides an overview of the raw data that we downloaded. The choice of sample periods is because of data availability, only starting after the introduction of the euro, and not choosing too dissimilar time periods (this is why we do not include the timespan 2019-2024 for yield curve data).

We should note that we have not seasonally adjusted any of the collected data ourselves. How-

[3] All data has been retreived on 24 March 2025.

[4] Note that the composition of the Eurozone has changed within our sample sizes, with for example Malta and Cyprus joining in 2008. For Eurozone data, we clarify if the composition has changed over time or if a fixed composition is used throughout the section.

ever, due to the availability of some variables only in seasonally adjusted format and the limited availability of non-seasonally adjusted data for others, the final dataset includes seasonally adjusted data. The reason for including mainly non-seasonally adjusted variables is that (i) we did not have a clear theoretical justification for using seasonally adjusted data, as it was not evident to us that such adjustments would lead to improved forecasts of exchange rate returns and since (ii) Rossi (2013), in a case study across various predictive exchange rate models using seasonally adjusted data, notes that 'empirical results based on seasonally unadjusted data are qualitatively similar'.

For the monthly USD/EUR exchange rate, we use an indicator of the Bank of International Settlements (BIS)[5] for the nominal exchange rate at the end of each month. We extract data from 1999-01(-31), corresponding to the introduction of the euro, to 2018-12(-31), which marks the endpoint of several other datasets extracted for analysis. We compute the log difference of this monthly data: each month's value is the logarithm of its end-of-month value minus the logarithm of the previous months' end-of-month value.

2.1 'Fundamental' Variables

First, we use a set of 'fundamental' variables capturing the economic and financial state within each month. We specify and discuss the variables used in this set below.

Consumer Price Index

As a measure of the price level in the Eurozone and the USA, we use monthly data for the Consumer Price Index (CPI) with base year 2015 from the OECD Data Explorer. Data is collected for the period January 1999 to December 2018 for the Eurozone and the US. More precisely, the estimate of the household consumption price level for the Eurozone is the Harmonised Index of Consumer Prices (HICP) from Eurostat, for the current composition of the Eurozone (20 countries). For the United States, CPI is calculated by the Bureau of Labor Statistics. For both samples, we compute the log difference by for each month taking the logarithm of that month's value minus that of the previous month's value.

Term Spreads

As a measure of expectations of long-term interest rates and macroeconomic conditions, we use data for the 10-year ('long-term') and 3-month ('short-term') monthly average interest rates from the OECD Data Explorer[6] for January 1999 to December 2018. The data for the Eurozone originates from the European Central Bank and uses the evolving composition of the Eurozone.[7]

[5]Note that, contrary to our definition, the BIS defines the US dollar in terms of euros 'EUR/USD': referring to how much euro to pay or receive for each US dollar, instead of the price of the US dollar divided by the price of the euro.

[6]More details on the computation of this data is available at the linked datasets.

[7]According to the OECD Data Explorer, which notes: Data prior to 2001 refers to EU11 (Belgium, Germany, Ireland, Spain, France, Italy, Luxembourg, the Netherlands, Austria, Portugal and Finland). Data from 2001 to 2006 refers to EU12 (EU11 plus Greece). Starting in January 2007, data refers to EU13 (EU12 plus Slovenia). Data refers to EA19 with Malta and Cyprus joining the euro area in 2008, Slovak Republic in 2009, Estonia in 2011, Latvia in 2014 and Lithuania in 2015.

The data for the United States originates from the Federal Reserve Board of the US. For both samples, we compute the term spread as the 10-year minus the 3-month interest rate.

National Financial Conditions Index (NFCI)

As a measure of the overall financial (and economic) state of the US, we use the weekly (ending Friday) National Financial Conditions Index (NFCI) of the Federal Reserve bank of Chicago, collected from the FRED from 1 January 1999 to 28 December 2018. As noted by the FRED, this data is based on money, debt, and equity markets and traditional as well as 'shadow' banking systems. Positive values indicate tighter than average financial conditions, while negative values indicate looser than average financial conditions Federal Reserve Bank of Chicago (n.d.). To use the index at our monthly frequency, we select the index of the last week in that month for which Friday is contained in that month.

Industrial Production

We use a seasonally adjusted monthly index for the industrial production (with as a base year 2010) in the Eurozone and the United States. The data is collected from the International Financial Statistics (IFS) dataset published by the International Monetary Fund (IMF), from January 1999 to April 2018 (no Eurozone data is available in the dataset after April 2018). The index is compiled using the methodology outlined by the United Nations Statistics Division (2010). For both samples, we compute the log difference by for each month taking the logarithm of that month's value minus that of the previous month's value.

Gross Domestic Product

As a measure of overall output, we also use seasonally adjusted quarterly data from the FRED of the Gross Domestic Product (GDP) for the Eurozone and the United States, from 2000-Q1 to 2018-Q3. For both samples, we compute the quarterly log difference as the logarithm of GDP in each quarter minus the logarithm of the GDP in each quarter's previous quarter. To use this log difference at our monthly frequency, we set the log difference GDP value for each month as that of the most recently finished quarter before that month[8], to prevent the use of future information in our one-month-ahead forecasting framework.

Debt-to-GDP ratio

As a measure of government borrowing, we use quarterly data for the percentage Debt-to-GDP ratio from 2000-Q1 to 2018-Q3. The data is collected from the ECB Data Portal for the Eurozone, using a fixed composition of the current 20 Eurozone countries. Any pre-euro domestic currencies are converted using fixed parity and expressed as a ratio of the annual moving sum of gross domestic product European Central Bank (2025). For the United States, we use seasonally adjusted data obtained from the FRED. We were inspired to use the Debt-to-GDP ratio by Cayen et al. (2010), who finds by use of a dynamic two-factor model that the

[8] i.e. Q1 for April, May, and June; Q2 for July, August, September; Q3 for October, November, December; Q4 for January, February, March.

US debt-to-GDP ratio was the first of two common factors that had a statistically significant cointegration relationship with the exchange rate in the long run.

For both the Eurozone and the United States, we compute its quarterly log difference as the logarithm of debt-to-GDP ratio in each quarter minus the logarithm of the debt-to-GDP ratio of the quarter before it. To use this log difference at our monthly frequency, we then set the log difference debt-to-GDP ratio for each month as that of the most recently finished quarter before that month[9], to again prevent the usage of future information in forecasting.

Net Current Account

As a measure of (changes in) trade, we collect the total net current account in goods and services, measured in euros for the Eurozone and in US dollars for the US. The data is obtained from the International Financial Statistics (IFS)[10] dataset published by the International Monetary Fund (IMF), ranging from 1999-Q1 to 2018-Q4.

When a country is exposed to a current account deficit, the traditional macro-economic theory suggests that, in the long-term, a country should produce a trade surplus to restore the balance in the current account. However, Gourinchas & Rey (2007) propose that part of the adjustment may occur via a wealth transfer between the country and the rest of the world, induced by a depreciation of the domestic currency. Because of this correction, they argue that the current account can be an important predictor of future exchange rates. Inspired by this paper, we include this measure in our dataset.

For each quarter, we calculate the difference between the United States and the Eurozone by subtracting the Eurozone net current account from the United States net current account. To use this difference at our monthly frequency, we use for each month that value of the most recently finished quarter before that month.[11]

2.2 'Experimental' Variables

Second, we use a set of 'experimental' variables. To the best of our knowledge, these variables have been relatively underexplored in the academic literature for improving exchange rate (density) forecasts.

S&P 500

Firstly, we use the US equity index S&P 500, including 500 large-cap listed U.S. equities S&P Dow Jones Indices (n.d.). We obtain the monthly average of the close price from the ECB Data Portal, for 1999-01-01 to 2018-12-31. We consider this equity index as it incorporates expectations about the state of the US economy, and because Bahmani-Oskooee & Sohrabian (1992) found a bidirectional causality between the S&P 500 index and the effective exchange rate

[9] Idem as for the Debt-to-GDP ratio.

[10] The updated data portal provides net current account data only at an annual frequency. However, the previous version of the portal continues to offer net current account data at a quarterly frequency.

[11] Idem as for the net current account.

of the dollar in the short-run. For each month, we compute the log difference as the logarithm of that month's price minus the logarithm of the previous month's price.

EURO STOXX 50

Secondly, we use the EURO STOXX 50, composed of leading blue-chip equities in the Eurozone. We obtain the monthly average of the close price from the ECB Data Portal, for the sample of 1999-01-01 to 2018-12-31. We include the index for the same reasons as we include the S&P500 index, but then for specific application within the Eurozone. For each month, we compute the log difference as the logarithm of that month's price minus the logarithm of the previous month's price.

Oil Prices

Thirdly, we obtain the daily West Texas Intermediate (WTI) crude oil prices from the FRED for the range of 1999-01-04 to 2018-12-31. We consider oil prices as S.-S. Chen & Chen (2007) provide empirical evidence for a cointegration relationship between real crude oil prices and real exchange rates. To use this daily data at our monthly frequency, we select the oil price of the last day in the month for which the data is reported (some days are missing, and some days have no data). Furthermore, we compute the log difference, for each month the logarithm of the last-available-day-of-the-month oil price minus the logarithm of that of the previous month.

2.3 Yield Curve Data

Finally, we also consider data on the yield curve. For the Eurozone, we use zero-coupon yield curve data for euro area central government bonds, and zero-coupon yield curve data aggregated for all 'AAA-rated' euro area central government bonds, downloaded from the ECB. [12] Both datasets are estimated by the ECB using the Svensson model[13], starting from 2004-09-07 and ending on 2024-12-31. From this full dataset, we extract the spot rates for the following maturities: 3-Month (SR_3M), 6-Month (SR_6M), 1-Year (SR_1Y), 2-Year (SR_2Y), 3-Year (SR_3Y), 5-Year (SR_5Y), 7-Year (SR_7Y), 10-Year (SR_10Y), 20-Year (SR_20Y), 30-Year (SR_30Y).

For the US, we collect the 'Market Yield on U.S. Treasury Securities at Constant Maturity, quoted on an Investment Basis' from 2004-09-07 to 2024-12-31 from the FRED for the following maturities: 3-Month, 6-Month, 1-Year, 2-Year, 3-Year, 5-Year, 7-Year, 10-Year, 20-Year, and 30-Year. Note that this is par rate (and not spot rate) data. It is estimated using a 'quasi-cubic hermite spline function' using 'bid-side market quotations' U.S. Department of the Treasury (n.d.).

[12] Both are under 'Downloads (SDMX format)' for all-rates under 'All years - euro area' and the AAA-rates under 'All years - AAA'. The CSV files have been downloaded for analysis. Note that each of these raw data files is substantial in size, exceeding 3 GB.

[13] See the 'yield curve technical notes file'.

Table 1: Overview raw data

Variable name	Description
Raw data for (1) Fundamental and (3) Composite forecasting index	
Monthly Raw Data	
Consumer Price Index, Eurozone	Price level estimate, Eurozone
Consumer Price Index, USA	Price level estimate, USA
10-year interest rate, Eurozone	Long-term interest rates, Eurozone
10-year interest rate, USA	Long-term interest rates, USA
3-month interest rate, Eurozone	Short-term interest rate, Eurozone
3-month interest rate, USA	Short-term interest rate, USA
NFCI	National Financial Conditions Index
Industrial Production, Eurozone	Total industrial output of the Eurozone
Industrial Production, USA	Total industrial output of the USA
Quarterly Raw Data	
GDP, Eurozone	Gross Domestic Product
GDP, USA	Gross Domestic Product
Debt-to-GDP, Eurozone	Ratio of government debt to total GDP
Debt-to-GDP, USA	Ratio of government debt to total GDP
Net Current Account, Eurozone	Total Net Current Account
Net Current Account, USA	Total Net Current Account
Raw data for (2) Experimental and (3) Composite forecasting index	
WTI Crude Oil Index	Benchmark for prices in the oil markets
S&P 500	Stock market index, USA
EURO STOXX 50	Stock market index, Eurozone
Raw data for (4, 5, 6, 7) Relative Yield Forecasting indices	
USA par rate yields: 3 and 6 month, and 1, 3, 5, 7, 10, 20, 30 year	Market Yield on U.S. Treasury Securities at Constant Maturity, Quoted on an Investment Basis
Eurozone spot rate yields, AAA	Yields 'AAA-rated' Eurozone central government bonds
Eurozone spot rate yields, ALL	Yields ('all') Eurozone central government bonds

3 Methodology

3.1 Basic Methodological Framework

Before describing our methodology, it should first be considered in which manner exchange rates should be studied. A perhaps intuitive approach would be to simply model the h-period ahead conditional distribution of the 'absolute' value of an exchange rate. However, Wang & Wu (2012) show that almost any forecasting model in the exchange rate literature rather focuses on the growth rate of exchange rates (i.e., log returns). We decide to continue with this approach.

Before further specifying the methodology, we first formally introduce a model set-up for fore-

casting using conditional quantiles. We focus on the one-month horizon (h = 1) in modeling future exchange rates. The model of interest in this paper takes the form:

$$s_{t+1} - s_t = x_t' \beta_\tau + \epsilon_{t+1} \tag{1}$$

where s_t is the logarithm of the nominal USD/EUR exchange rate (the price of one US dollar in euros) at time t, $s_{t+1} - s_t$ is the return of the 1-month-ahead nominal exchange rate at time t, x_t is a n x 1 vector of explanatory variables, and ϵ_{t+1} is an error term. β_τ is defined as a 1 x n vector of coefficients that depends on the τ-th quantile of ϵ_{t+1}. The τ-th quantile of ϵ_{t+1} is the value q_τ for which it holds that $P(\epsilon_{t+1} < q_\tau) = \tau$.

In this paper, we define the vector of explanatory variables x_t to contain a constant, a one-period lag of log exchange rate returns ($s_t - s_{t-1}$), and a single predictive index value. Note that Equation 1 suggests a level forecast of the exchange rate returns. However, the objective of this paper is to analyze the entire distribution of $s_{t+1} - s_t$ around this point forecast. Also note that we are creating a *conditional* density forecast, as we allow x_t to contain any variables that may improve the quality of the forecast. An unconditional density forecast would be computed by restricting x_t to only contain a constant.

3.2 Predictive Index

In this section, we introduce the various predictive indices that we consider as conditional variables. Table 2 provides an overview of the predictive indices. Besides, an overview of all transformed data used to construct the predictive indices is available in Appendix B. Due to the limited number of observations and the computational cost of the estimation procedure, we restrict x_t to contain only a single predictive index at a time, even though this may limit predictive power. However, it also makes interpretation more straightforward throughout this paper.

PCA-based Predictive Index

The first set of indices that we use is based on the Principal Component Analysis of the standardized[14] data of the entire sample. For each forecasted month, the predictive index is the scores of the standardized data of the previous month on the first principal component. Note that our forecasting method using the PCA-based predictive index will thus use the past, i.e. data from the previous month, but in practice this data may not always be easily available at the start of the month for which one forecasts the exchange rate. As the focus of our research is to investigate whether this technique can help make good density forecasts at all, we are generous and did not restrict to variables which can be obtained or now-casted accurately or easily for practical forecasts of one-month-ahead exchange rate returns. In addition, note that the Principal Component Analysis is performed on the whole sample: while the score for each month is determined by the size of the variables in the previous month, the axis of the first principal component is determined by the full dataset, which in practice would not be possible

[14] Such that each variable has a mean 0 and variance 1 across time.

in real time. However, estimating this principal component on the 20 preceding years of our current sample presents the issue that the economy and monetary policy have likely been substantially different, and that the euro did not exist yet. Alternatively, the use of dynamic PCA as introduced by Koop & Korobilis (2014) appeared initially promising. However, given the limited number of monthly observations, applying dynamic PCA to a short time window could overly emphasize specific changes in variables, such as an increasing oil price. Additionally, it is hardly interpretable across time and for the in-sample analysis.

Our first index named 'Fundamental Forecasting Index (FFI)' is the first principal component of (1) the log difference of the CPI in the Eurozone; (2) the log difference of the CPI in the US minus the log difference of the CPI in the Eurozone; (3) the term spread of the US; (4) the term spread of the Eurozone; (5) the NFCI; (6) the log difference of the industrial production in the US; (7) the log difference of the industrial production in the Eurozone; (8) the log difference of the GDP in the US; (9) the log difference of the GDP in the Eurozone; (10) the log difference of the debt-to-GDP ratio in the US; (11) the log difference of the debt-to-GDP ratio in the Eurozone; and (12) the total net current account of the USA minus the total net current account of the Eurozone. Our second index named 'Experimental Forecasting Index (EFI)' is the first principal component of (1) the log difference of the S&P500; (2) the log difference of the EURO STOXX 50; (3) the log difference of the WTI Oil Price index. Our third index named 'Composite Forecasting Index (CFI)' is the first principal component of all variables used to construct the above two indices FFI and EFI.

Relative Nelson-Siegel Predictive Index

The yield curve contains information about market expectations of the future, and bond price data is readily available, as these bonds are actively traded. Our second set of predictive indices is thus inspired by Y.-C. Chen & Tsang (2013), who use Nelson-Siegel factors on the yield spread between the United States and Canada, Japan, and the United Kingdom to make statistically significant point forecasts for the exchange rate of their currencies relative to the US dollar. As Y.-C. Chen & Tsang (2013) note, the literature has found that a higher level factor indicates a higher long-term inflation expectation. As this captures expectations about the economy and the relative value of the currency, we postulate it might be informative for our USD/EUR density forecasts, and use it as a predictive index. Furthermore, Y.-C. Chen & Tsang (2013) note that the literature found that the slope factor captures expectations about the business cycle, with a higher slope factor value meaning a flatter slope captures an expected output decrease. By the construction of the Nelson-Siegel model, the slope factor weighs the short term more. As we aim to forecast the one-month-ahead density forecast, these short term expectations, and expectations about a possible relative slowdown, we consider the slope factor as well.

To compute these indices for each month, we take the yield of Eurozone central government bonds minus the yield of US Treasuries of the last day of the previous month. Similar to Y.-C. Chen & Tsang (2013), for each month t we then estimate the level, slope, and curvature factors, (L_t^R, S_t^R, C_t^R respectively), using the following Nelson-Siegel OLS regression on these

'relative yields' for different maturities m:

$$i^m_{t,\ Eurozone} - i^m_{t,\ US} = L^R_t + S^R_t \left(\frac{1-e^{-\lambda m}}{\lambda m}\right) + C^R_t \left(\frac{1-e^{-\lambda m}}{\lambda m} - e^{-\lambda m}\right) + \epsilon^m_t \tag{2}$$

with $i^m_{t,\ Eurozone}$ the Eurozone and $i^m_{t,\ US}$ the US yield that month, ϵ^m_t the fitting error term, and λ set[15] to 0.0609.

This results in the predictive index (4) $LFAAA$, the level factor using AAA-rated Eurozone bonds; (5) $LFALL$, the level factor using all Eurozone bonds; (6) $SFAAA$, the slope factor using AAA-rated Eurozone bonds; (7) $SFALL$, the slope factor using all Eurozone bonds. We consider both the Eurozone AAA-rated central government bond yields as well as all Eurozone central government bond yields, as these different central government yields are a particular feature of the Eurozone. We postulate they may capture different signals related to the state of the economy, the risks it faces, and the USD/EUR exchange rate.

Note that while Y.-C. Chen & Tsang (2013) use zero-coupon yields computed using the Fama-Bliss bootstrapping methodology, we instead use data for the Eurozone zero-coupon spot rates estimated using the Svensson model (an extension of the Nelson-Siegel model), and par yields[16] for the US estimated using another technique. This is a practical limitation of our research.

Table 2: Overview Predictive Indices

Model	Variable	Description
	PCA-based Predictive Indices	
1	FFI_t	Fundamental forecasting index
2	EFI_t	Experimental forecasting index
3	CFI_t	Composite forecasting index
	Relative Yield Nelson-Siegel Factor Predictive Indices	
4	$LFAAA_t$	Level Factor, Eurozone AAA-rated - US Treasury
5	$LFALL_t$	Level Factor, Eurozone (all) - US Treasury
6	$SFAAA_t$	Slope Factor, Eurozone AAA-rated - US Treasury
7	$SFALL_t$	Slope Factor, Eurozone (all) - US Treasury

3.3 Conditional Density Forecast

To create density forecasts for the USD/EUR nominal exchange rate returns, we will use the 2-step procedure introduced by Adrian et al. (2019b). The procedure creates a density forecast by using a quantile regression interpolation framework to fit a skewed-t distribution. However, it should first be noted that a restrictive assumption regularly made in the past, as defended by studies such as Coppes (1995), has been normality of exchange rate returns. However, a firm strand of the literature, including the work of de Vries (1994), argue that fatter and asymmetric tails are stylized facts of exchange rate returns. As the skewed-t distribution encompasses a

[15] As done by Y.-C. Chen & Tsang (2013), who note this is standard in the literature.

[16] Typically more smoothed than zero-coupon 'spot' yields, these capture the general tendency in the yield curve as well. It is a limitation of our research, or at least a difference with Y.-C. Chen & Tsang (2013).

variety of distributions, including the normal distribution itself, Adrian et al.'s technique allows for non-normality in density predictions of exchange rate returns. Our main motivation for choosing this approach is that it allows us almost surely to come closer to the true DGP of the predictive distribution.

Let y_{t+h} denote the h-month-ahead USD/EUR nominal exchange rate returns $s_{t+h} - s_t$, and let x_t be a vector containing explanatory variables at time t. In a quantile regression of y_{t+h} on x_t, the regression slope $\hat{\beta}_\tau$ is estimated by minimizing a weighed sum of absolute errors. Intuitively, this weighing forces an approximate proportion τ of the observations to be below the quantile regression fit and an approximate proportion $(1-\tau)$ of the observations to be above the quantile regression fit, resulting in an estimate of the quantile. More precisely, we estimate β_τ as:

$$\hat{\beta}_\tau = \underset{\beta_\tau \in R^k}{\arg\min} \sum_{t=1}^{T-h} (\tau \cdot \mathbf{1}(y_{t+h} \geq x_t\beta_\tau)|y_{t+h} - x_t\beta_\tau| + (1-\tau) \cdot \mathbf{1}(y_{t+h} < x_t\beta_\tau)|y_{t+h} - x_t\beta_\tau|) \quad (3)$$

where $\mathbf{1}(\cdot)$ is the indicator function that determines a specific weight on the absolute error term depending on whether this error term is above or below the quantile. The conditional quantile estimate of y_{t+h} on x_t can than be computed as:

$$\hat{Q}_{y_{t+h}|x_t}(\tau|x_t) = x_t\hat{\beta}_\tau \quad (4)$$

We perform this estimation for the 5%, 25%, 75% and 95% quantile levels. The resulting set of estimated conditional quantiles will then be used to fit a predictive distribution in the second step.

To smooth the obtained quantile functions into a probability density function, we fit a skewed-t distribution to the estimated quantiles. The probability density function (PDF) of the skewed-t distribution developed by Azzalini & Capitanio (2003) is given by:

$$f(y;\mu,\sigma,\alpha,\nu) = \frac{2}{\sigma} t\left(\frac{y-\mu}{\sigma};\nu\right) T\left(\alpha\frac{y-\mu}{\sigma}\sqrt{\frac{\nu+1}{\nu+(y-\mu)^2/\sigma^2}};\nu+1\right), \quad (5)$$

where $t(\cdot)$ and $T(\cdot)$ respectively denote the PDF and CDF of the Student t-distribution. The parameter μ may be interpreted as the location of the distribution, σ as the scale of the distribution, ν as the fatness of the distribution's tails and α as a shape parameter. The given skewed-t distribution provides a high level of flexibility and demonstrates the ability to capture asymmetry in financial data.

To estimate the skewed-t distribution's parameters $\{\mu_t, \sigma_t, \alpha_t, \nu_t\}$, we minimize the quadratic difference between the empirical quantile estimates obtained in Equation (4) and the quantile function of the skewed-t distribution, summed over the 5%, 25%, 75% and 95% quantile levels:

$$\{\hat{\mu}_{t+h}, \hat{\sigma}_{t+h}, \hat{\alpha}_{t+h}, \hat{\nu}_{t+h}\} = \underset{\mu,\sigma,\alpha,\nu}{\arg\min} \sum_\tau \left(\hat{Q}_{y_{t+h}|x_t}(\tau|x_t) - F^{-1}(\tau;\mu,\sigma,\alpha,\nu)\right)^2, \quad (6)$$

where $F^{-1}(\tau; \mu, \sigma, \alpha, \nu)$ is the inverse cumulative density function (CDF) of the skewed-t distribution.

Using the above described procedure, we will fit a skewed-t distribution for the one-month-ahead USD/EUR exchange rate returns conditional on a constant, the current log exchange rate returns $(s_t - s_{t-1})$, and a single predictive index value. Similar to the approach of Adrian et al. (2019b), we will also create unconditional density forecasts for the USD/EUR exchange rate returns. This is equivalent to applying the 2-step procedure to fit a skewed-t distribution for the one-month ahead USD/EUR exchange rate returns conditional on a constant.

To apply the method described above, we should motivate a scheme to use in out-of-sample forecasting. In practice, regularly considered schemes include rolling window estimation and recursive estimation. A rolling window is a favorable choice when the predictive relationship between the independent variable and dependent variables are time-varying. Intuitively, historical independent variables may partially or entirely lose relevance when predicting exchange rates further into the future. Besides, a rolling window allows the forecasting model to adapt more quickly to structural breaks in exchange rates. Rossi (2013) argues that the predictive ability of macro-economic and financial variables is time-varying. Clark (2011) argues that obtaining reliable density forecasts for macro-economic variables requires a method that can handle repeated structural breaks. Therefore, a reasonable approach is to continue with a rolling window estimation.

Following the notation of Yapi (2020), applying a rolling window estimation scheme as described above yields a set of out-of-sample density forecasts $\{\hat{f}_{t+h}(y_{t+h}|\mathcal{I}^t_{t-R+1})\}_{t=R}^T$, where y_{t+h} represents the h-month-ahead nominal exchange rate returns, $\hat{f}_{t+h}$ denotes a predicted skewed-t probability density function, $\mathcal{I}^t_{t-R+1}$ is the information set of all available data in the given window iteration and R is defined as the size of the estimation window. There is no clear consensus in the literature on the choice of R. A small window may result in an unreliable forecast, whilst a large window may result in an inability of the forecasting scheme to quickly adapt to structural breaks in the data. Due to the lack of consensus, this paper will investigate the monthly estimation windows $R = 24, 48, 72, 96$ (i.e., 2, 4, 6, and 8 years).

Finally, we should formally define a benchmark model to compare the introduced framework with. A widely used benchmark in the literature for density forecasting is the random walk model without drift, as suggested by Rossi (2013) and Hong et al. (2007). To generate one-month-ahead density forecasts from this model, we assume a normal distribution where the mean is equal to the current observed value of the exchange rate returns. The variance of the distribution is estimated using a rolling window of past forecast errors, with the window size matching that of the competing forecasting model. This approach follows the methodology of Gaglianone & Marins (2017).

3.4 Forecast Evaluation

We will conduct one-month-ahead out-of-sample density forecasts employing seven predictive index specifications and four different rolling window sizes. To assess out-of-sample forecasting

performance, we will include and conduct a calibration test and a comparative test.

Calibration Test

To evaluate the calibration of our out-of-sample density forecasts, we will conduct the test introduced by Rossi & Sekhposyan (2019). This statistic tests the null of correctly specified density forecasts given the estimation method and the chosen rolling window size. The test is an absolute test, which implies that it does not yield information on one model potentially outperforming another model. The choice for this test is in line with similar density forecasting literature (e.g. Yapi (2020) and Adrian et al. (2019b)) making use of the two-step quantile regression approach for forecasting.

The test computes the Probability Integral Transformations (PITs) from a sequence of out-of-sample forecasts and constructs their empirical cumulative distribution function (CDF). Diebold et al. (1998) show that if the predicted densities are correctly calibrated, the resulting PITs follow an independent and identically distributed Uniform(0,1) distribution with a CDF corresponding to a 45-degree line.

To formally describe the testing procedure, we follow the notation of Yapi (2020). The first step of the procedure is the computation of the PITs. Given a predicted skewed-t probability density function $\hat{f}_{t+h}$, the PIT is defined as the CDF of the density function evaluated at the realized value y:

$$w_{t+h} = \int_{-\infty}^{y_{t+h}} \hat{f}_{t+h}(y|\mathcal{I}_{t-R+1}^{t})\, dy = \hat{F}_{t+h}(y_{t+h}|\mathcal{I}_{t-R+1}^{t}) \tag{7}$$

Under correctly calibrated predictive density specifications, the resulting sequence $\{w_{t+h}\}$ follows an i.i.d. Uniform(0,1) distribution. The null hypothesis of correctly specified densities can formally be defined as:

$$H_0 : \hat{F}_{t+h}\left(y|\mathcal{I}_{t-R+1}^{t}\right) = \hat{F}_0\left(y|\mathcal{I}_t\right) \quad \text{for all } t = R, \ldots, T \tag{8}$$

where $\hat{F}_0\left(y|\mathcal{I}_t\right) = P(y_{t+h} \leq y|I_t)$ is the specified distribution under the null and I_t is the true information set.

Instead of applying the Kolmogorov-Smirnov (KS) test to evaluate the null hypothesis, we conduct a graphical analysis to examine possible deviations. The KS-test focuses on violations of the null within the entire forecasted distribution. A graphical analysis, however, provides the benefit of yielding insights into which sections of the distribution (i.e., left-tail, right-tail, or center) the predicted density deviates from the theoretical density. In the graphical analysis, the empirical CDF of $\{w_{t+h}\}$ will be compared to a 45-degree line. More specifically, the null will be significantly rejected if the empirical CDF breaches the 5% confidence bands, which are tabulated in Rossi & Sekhposyan (2019) and derived from Monte Carlo simulations.

Comparative Test

A common way to compare the relative performance of a pair of models is to compute the 'logarithmic scores' of each model's density forecasts over time. The score can be formally defined as $\log \hat{f}_t(y_{t+1})$, where $\hat{f}_t$ denotes the predicted skewed-t distribution at time t and y_{t+1} is the realized value of the exchange rate return at time $t+1$. A higher score is favorable, since this implies that the density function assigns a higher probability mass to the observed outcome.

In our paper, we apply the framework of Amisano & Giacomini (2007), following the approach of Adrian et al. (2019b) and Gaglianone & Marins (2017), to formally test if our two-step quantile regression approach outperforms the random walk in out-of-sample density forecasting. As defined in Equation (10), one can test the null hypothesis of an average difference of zero between the log-scores of two sets of density forecasts. In our case, this involves comparing the random walk model to our various model specifications using the Log Predictive Scores Difference (LPSD), as shown in Equation (9). Given a density forecast $\hat{f}_t^{RW}$ and an alternative density forecast $\hat{f}_t$:

$$LPSD_{t+1} = \log \hat{f}_t^{RW}(y_{t+1}) - \log \hat{f}_t(y_{t+1}) \tag{9}$$

$$H_0 : E[LPSD_{t+1}] = 0 \tag{10}$$

Consequently, the null hypothesis can be formally evaluated according to the following test statistic:

$$t = \frac{\overline{LPSD}}{\hat{\sigma}_n^{HAC}/\sqrt{n}} \tag{11}$$

with the numerator denoting the average LPSD, i.e. $\overline{LPSD} = \frac{1}{P}\sum_{t=R}^{T-1} LPSD_{t+1}$ and P denoting the number of out-of-sample forecasts conducted. Furthermore, the estimator for the asymptotic variance of the LPSD is heteroskedasticity and autocorrelation consistent (HAC) and can be defined as $\hat{\sigma}_n^{HAC} = Var(\sqrt{n} * \overline{LPSD})$. A negative test statistic indicates that the two-step quantile approach provides superior performance in terms of the LPSD relative to a random walk model.

Note that we do make use of the unweighted variation of the test, since a collection of papers including Gneiting & Ranjan (2011) and Diks et al. (2011) noted that the scoring rule of the weighted variation is improper. That is, an incorrectly specified density forecast could score better than the true model due to the scoring rule favoring densities with relatively more probability mass in the region of interest (e.g. fat tails).

3.5 Tail analysis Methodology

We will assess and discuss tail-risk using two different metrics in this paper. First, we examine the Expected Shortfall and its right-tail counterpart, the Expected Longrise, based on our predicted conditional densities. Secondly, we conduct a comparative analysis of relative upside and downside entropy in exchange rate returns, focusing on differences between the unconditional

and conditional distributions.

The Expected Shortfall and Longrise give us more insight into the absolute values associated with extreme events. The Expected Shortfall is defined as the conditional expectation of losses exceeding the Value-at-Risk (VaR) as described in Acerbi & Tasche (2002), while the Expected Longrise serves as its right-tail counterpart, capturing the expected gain beyond the corresponding upper quantile. In the case of exchange rate returns, this can be interpreted as the depreciation and appreciation risk for the domestic currency, respectively. Formally, they are defined as:

$$ES_{t+h} = \frac{1}{\pi} \int_0^{\pi} F^{-1}_{y_{t+h}|x_t}(\tau \mid x_t)\, d\tau \tag{12}$$

$$EL_{t+h} = \frac{1}{\pi} \int_{1-\pi}^{1} F^{-1}_{y_{t+h}|x_t}(\tau \mid x_t)\, d\tau \tag{13}$$

where π denotes the selected target probability as a fraction (e.g., 0.05 implies 5%) and $F^{-1}_{y_{t+h}|x_t}(\tau|x_t)\, d\tau$ denotes the conditional forecasted CDF.

The second metric considered is the relative downside entropy and its right-tail counterpart, the relative upside entropy. They represent the excess risk that the conditional distribution assigns to the respective side of the distribution compared to the unconditional distribution. Specifically, they measure how much more (or less) widely distributed or heavy-tailed the conditional density is on either side of the median forecast compared to the unconditional distribution.

We follow the notation of Adrian et al. (2019b), where relative downside entropy ($\mathcal{L}_t^D$) and relative upside entropy ($\mathcal{L}_t^U$) are formally defined as:

$$\mathcal{L}_t^D\left(\hat{f}_{y_{t+h}|x_t}; \hat{g}_{y_{t+h}}\right) = -\int_{-\infty}^{\hat{F}^{-1}_{y_{t+h}|x_t}(0.5|x_t)} \left[\log \hat{g}_{y_{t+h}}(y) - \log \hat{f}_{y_{t+h}|x_t}(y|x_t)\right] \hat{f}_{y_{t+h}|x_t}(y|x_t)\, dy \tag{14}$$

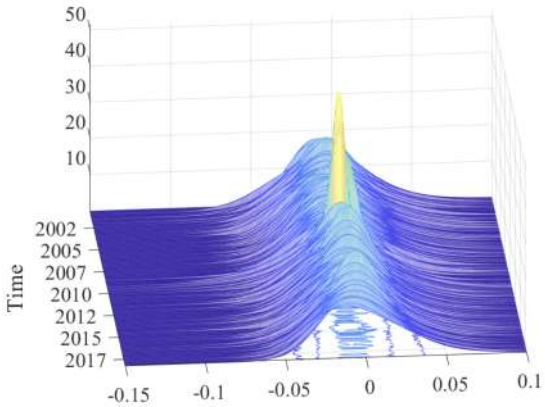
$$\mathcal{L}_t^U\left(\hat{f}_{y_{t+h}|x_t}; \hat{g}_{y_{t+h}}\right) = -\int_{\hat{F}^{-1}_{y_{t+h}|x_t}(0.5|x_t)}^{\infty} \left[\log \hat{g}_{y_{t+h}}(y) - \log \hat{f}_{y_{t+h}|x_t}(y|x_t)\right] \hat{f}_{y_{t+h}|x_t}(y|x_t)\, dy \tag{15}$$

where $\hat{g}_{y_{t+h}}$ is defined as the predicted unconditional density function, $\hat{F}_{y_{t+h}|x_t}(y|x_t)$ is the cumulative distribution function of the predicted conditional density function $\hat{f}_{y_{t+h}|x_t}(y|x_t)$ and $F^{-1}_{y_{t+h}|x_t}(0.5|x_t)$ is the conditional median.

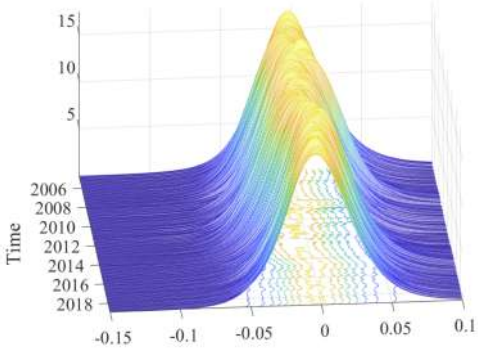
The values of the relative entropy can be interpreted as follows. A score of zero indicates that the unconditional and conditional distributions overlap exactly. A positive value indicates that the conditional distribution assigns more probability mass to extreme outcomes in the respective side of the distribution. A negative value suggests the opposite, i.e., the conditional distribution places less emphasis on extremes compared to the unconditional distribution. Note that the sum of these relative entropies is strictly non-negative due to the definition of relative entropy

given by Kullback & Leibler (1951).

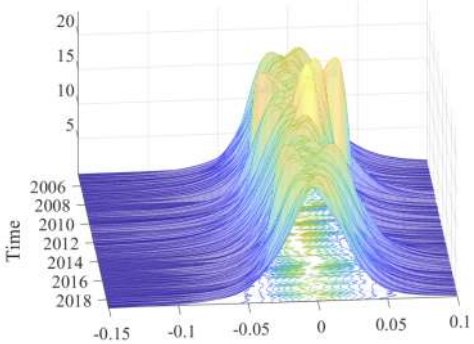
4 In-Sample Analysis: Quantile Regression Framework



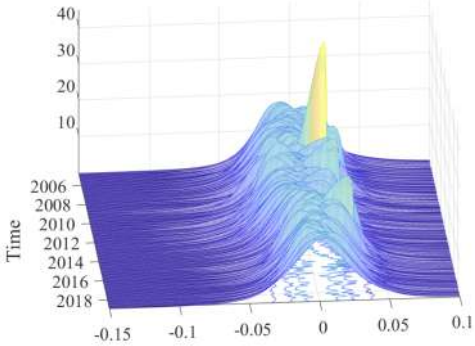
(a) CFI



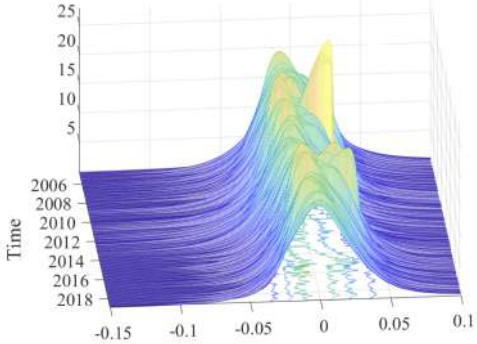
(b) LF AAA



(c) LF ALL



(d) SF AAA



(e) SF ALL

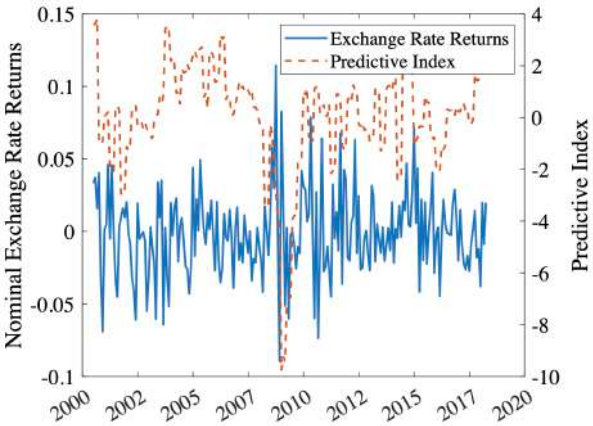
Figure 1: In-sample conditional distribution of USD/EUR exchange rate log return over time

Note: One-month-ahead model fit distribution of USD/EUR exchange rate returns conditional on a constant, the current log return of the nominal exchange rate and the noted predictive index.

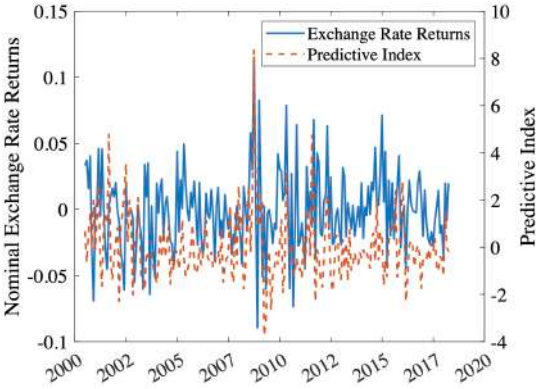
As shown in Figure 1, the estimated one-month-ahead conditional distributions of the USD/EUR exchange rate returns vary substantially over time, for various predictive indices[17]. For the CFI in Figure 1a, two features stand out in particular. First, the 2008 financial crisis is marked by a pronounced spike in the forecasted conditional density. During this period, the composite index

[17]Those for the FFI and EFI are broadly similar to CFI, and can be found in Appendix C.1.

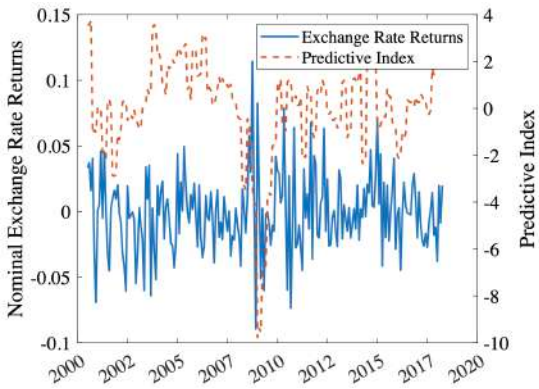
reflects heightened financial stress and elevated volatility. The quantile regression framework responds by fitting a heavy-tailed distribution to account for the increased likelihood of extreme returns. However, because a higher kurtosis also implies a sharper peak at the center, the model simultaneously places more mass near zero. This reflects not only tail risk but also the model's uncertainty about the direction of movement, leading to a density that is both fat-tailed and sharply peaked. Second, while the shape and width of the distribution remain relatively stable and symmetric during calmer periods, particularly before 2005 and after 2012, clear asymmetry emerges during crisis episodes. Specifically, the left tail exhibits more variation over time than the right tail does, suggesting that downside risk (dollar depreciation) fluctuates more strongly than upside risk in one-month-ahead exchange rate returns. The conditional distributions based on the model using relative Nelson-Siegel yield factors as predictive index, also vary over time. However, they model different distributions: those using the slope factor in Figure 1d and Figure 1e modeling much more asymmetric distributions, and those using the level factor in Figure 1b and Figure 1c much less peakedness for the 2008 financial crisis.



(a) FFI



(b) EFI



(c) CFI

Figure 2: Time series of forecasting indices (FFI, EFI, CFI) and nominal USD/EUR exchange rate log return

Figure 2 presents the time series of the Fundamental Forecasting Index (FFI), the Experimental Forecasting Index (EFI), and the Composite Forecasting Index (CFI) alongside monthly USD/EUR exchange rate returns. The returns reflect changes in the USD/EUR exchange rate, with positive values indicating a strengthening U.S. dollar (euro depreciation) and negative

values indicating a weakening dollar (euro appreciation)

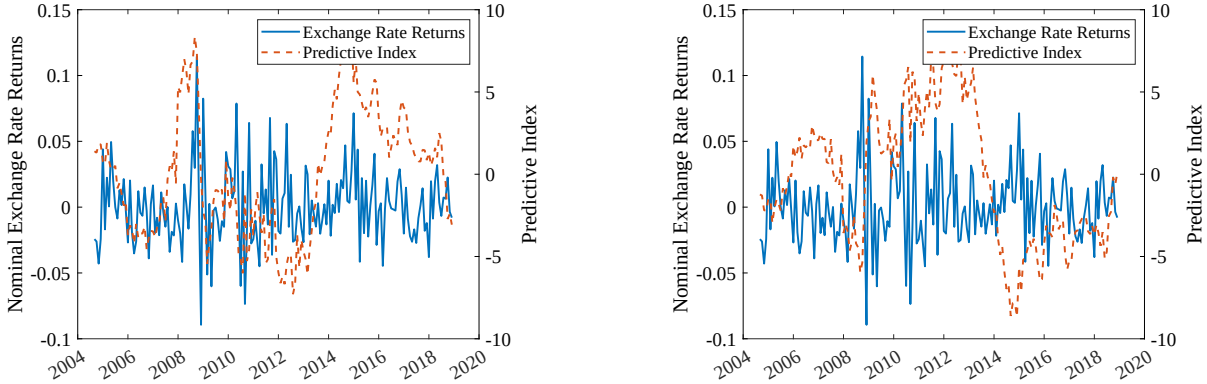
The FFI, shown in Figure 2a, moves more gradually than the exchange rate returns. Despite the difference in fluctuations, several turning points in the FFI seem to align with major shifts in the exchange rate returns. Furthermore, a particularly sharp drop during the Global Financial Crisis (2007–2008) ocurrs alongside increased return volatility. After 2009, the FFI rises again, aligning with dollar strength and decreasing volatility, and then stabilizes post-2010, a phase also marked by reduced volatility. Around 2015, a sharp downward spike in the FFI is followed by short-lived volatility, before both the index and returns stabilize again.

The EFI, displayed in Figure 2b, displays more fluctuating and abrupt movements compared to the smoother dynamics of the FFI. Rather than capturing underlying economic trends, the EFI appears more sensitive to short-term fluctuations in investor sentiment or external shocks. For instance, the pronounced spike during the Global Financial Crisis (2007–2008) suggests a rapid response to heightened financial instability. A similarly sharp surge around 2016 likely reflects a brief period of market optimism or risk repricing. However, these movements tend to be short-lived and less persistent than those observed in the FFI, indicating that the EFI may be better suited for signaling short-term volatility rather than sustained directional shifts in the exchange rate returns.

Finally, the CFI, shown in Figure 2c, combines both sets of variables from the FFI and EFI. Interestingly, it bears a strong visual resemblance to the FFI, particularly in its level and turning points. This suggests that fundamental variables dominate in the Principal Component Analysis, therefore carrying a larger weight in the composite index. Nevertheless, the inclusion of the experimental variables may still improve the signal, especially around more abrupt shifts in returns, such as those observed during crises periods.

Overall, while the three indices differ in construction, they each show some alignment with broader shifts in exchange rate dynamics. In particular, the FFI and CFI appear to capture medium-term trends, while the EFI adds a possible layer of explanation to short-term market movements. This supports the potential use of these indices as conditional variables for forecasting the distribution of future exchange rate changes.

Figure 3 presents the time series of the level factor of the Eurozone (all) - US treasury bonds (LF ALL) and the slope factor of the Eurozone (all) - US treasury bonds (SF ALL) alongside monthly USD/EUR exchange rate returns. The results for the level and slope factor of the Eurozone AAA-rated - US Treasury bonds are moved to Appendix C.2, as they are nearly identical to those of the level and slope factor of the Eurozone (all) - US Treasury bonds. The level factor depicted in Figure 3a, which past literature has linked to long-term inflation expectations, visually aligns with medium-horizon trends in exchange rate returns. A pronounced peak in the level factor from 2008 to 2009 is accompanied by a substantial spike in exchange rate returns, suggesting that market expectations about inflation and macroeconomic stability were reflected through volatile currency movements during the global financial crisis.



(a) LF ALL

(b) SF ALL

Figure 3: Time series of forecasting indices (LF ALL, SF ALL) and nominal USD/EUR exchange rate log return

The slope factor displayed in Figure 3b, often associated with business cycle expectations and short-term exchange rate dynamics, appears to show patterns that coincide with turning points in the exchange rate returns, notably around the 2008-crisis and the post-2015 period. Taken together, the figures offer visual indications that yield curve components contain valuable forward-looking information relevant to the prediction of exchange rate movements.

In Figure 4, we present a scatter plot of the one-month-ahead exchange rate returns against different forecasting indices. The scatter plot includes 5%, 50% and 95% univariate quantile regression lines, as well as an univariate OLS estimate. All regression outcomes result from an estimation on the entire available sample. We include results for the Fundamental Index (FFI_t), the Experimental Index (EFI_t), the level factor of the Eurozone AAA-rated - US Treasury bonds ($LFAAA_t$) and the slope factor of the Eurozone (all) - US Treasury bonds ($SFALL_t$). The results for the Composite Index (CFI_t) index are moved to Appendix C.3, as they are nearly identical to those of the Fundamental Index. To keep the analysis comprehensive, we also move the results of the level factor of the Eurozone (all) - US Treasury bonds ($LFALL_t$) and the slope factor of the Eurozone AAA-rated - US Treasury bonds ($SFAAA_t$) to Appendix C.3.

For the fundamental index, all regression lines except the flat 50% quantile regression line, slope upwards. This indicates that an increase in the fundamental index corresponds to macroeconomic conditions that support an appreciation of the dollar relative to the euro in the next month. The estimated slope of the 5% quantile regression line differs from the other quantile regression estimates in that its slope is steeper. This suggests that extreme returns of the USD/EUR exchange rate in the lower tails are more sensitive to movements in the index that captures the underlying macroeconomic fundamentals. The 5% and 95% quantile regression lines also suggest that the macroeconomic factors captured by our index appear to have a more pronounced influence on significant dollar depreciations relative to the euro, than on euro depreciations relative to the dollar.

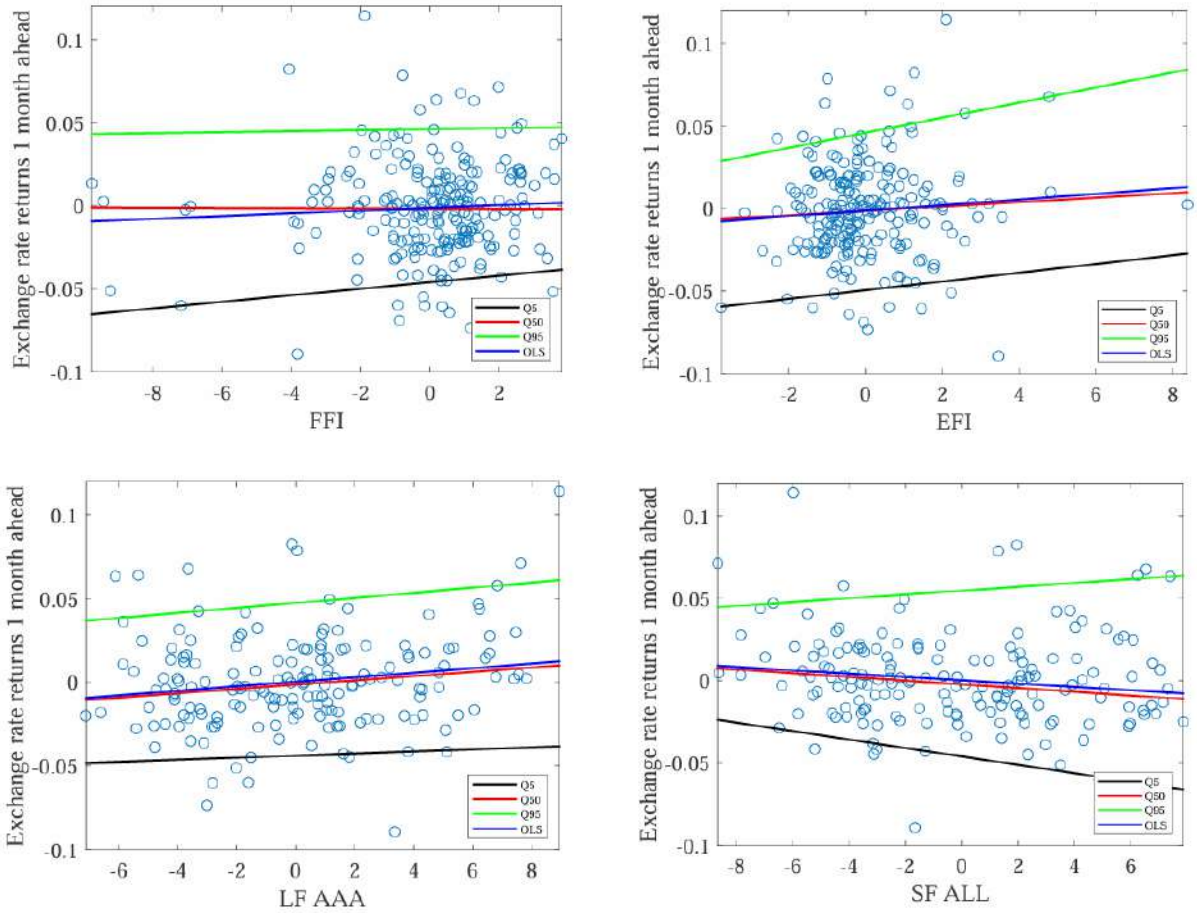


Figure 4: In-sample univariate quantile regressions of the one-month-ahead log exchange rate returns on the fundamental index, the experimental index, the level factor of the Eurozone AAA-rated - US Treasury bonds $LFAAA_t$ and the slope factor of the Eurozone (all) - US Treasury bonds $SFALL_t$

In these univariate regressions, the experimental index exhibits a positive relationship with the one-month ahead USD/EUR exchange rate return, as indicated by the positive slope of all estimated regression lines. An increase in the experimental index suggests oil-market and stock-market conditions that support a one-month ahead appreciation of the dollar relative to the euro. Alternatively, an increase in the experimental index can be interpreted as a strengthening of market sentiment towards the US as a perceived safe haven for investments in both the oil and large-cap stock markets. Consequently, this perception is associated with a subsequent appreciation of the dollar relative to the euro. The 95% quantile regression line is steeper than its 5% counterpart, suggesting that the underlying market conditions captured by the experimental index have a more substantial effect on extreme returns in the upper tails than in the lower tails.

An increase in the level factor $LFAAA_t$ can be interpreted as a widening of the yield spread between Eurozone AAA-rated bonds and US Treasury bonds. An increase in the level factor signals that investors perceive higher risks within the Eurozone relative to the US, leading them to demand a greater yield premium for Eurozone bonds. Furthermore, a larger spread indicates that investors expect the long-run inflation in the Eurozone to be higher than that of the US.

For all estimated univariate regression lines, an increase in the yield spread is associated with a one-month ahead appreciation of the dollar relative to the euro. A possible explanation for this relationship is that risk-averse investors, in search of the safest assets, may increasingly shift their capital towards US Treasuries when AAA-rated Eurozone bonds offer higher risk premia. This flight to safety could lead to a stronger demand for USD-denominated assets, thereby forcing an appreciation of the dollar relative to the euro.

An increased slope factor $SFALL_t$ captures market expectations of higher economic growth in the Eurozone relative to the US. The 5% and 50% quantile regression lines, as well as the OLS estimate, have a negative slope. This indicates that an increase in the slope factor, associated with a higher economic growth expectation in the Eurozone, is associated with a one-month ahead depreciation of the dollar relative to the euro. Meanwhile, the slightly increasing 95% quantile regression line suggests this higher slope factor is associated with a higher uncertainty in general, a potential for dollar appreciation in addition to depreciation.

5 Out-of-sample Forecasting and Testing

5.1 Out-Of-Sample Results

We will assess the calibration of our out-of-sample forecasts according to Rossi & Sekhposyan (2019), as described in our methodology. Results for the PITs are presented as figures in Appendix D. A perfectly calibrated model would correspond to the plotted 45-degree line, since this implies that the assigned probability below each quantile $Q_{y_{t+h|x_t}}(\tau)$ is equivalent to the theoretical fraction τ. Hence, calibration improves as the empirical CDF aligns more closely with this line. The figures include confidence bands at the 5%-level parallel to the 45-degree line following Rossi & Sekhposyan (2019). We present test results for the unconditional distribution and all specifications of the conditional distribution of the USD/EUR exchange rate returns.

For the 24-month rolling window size, all four yield curve-based specifications fall outside the confidence bands of the empirical CDF, thereby rejecting the null hypothesis of a correct density specification. For all larger window sizes, the empirical CDFs of the PITs are within the confidence bands. Therefore, we do not reject the null hypothesis for any of the remaining conditional distribution specifications. We can thus conclude that our densities are calibrated properly with the exception of the 24-month rolling window yield curve specifications.

A closer examination of the PITs reveals that, for the figures employing the 24-month rolling window, the conditional densities appear to have assigned more probability mass to the left tail of the distribution compared to the theoretical density. Conversely, the right tail exhibits an underestimation of the probability mass in case of all seven specifications. For the 48-month rolling window, the yield curve specifications still exhibit this pattern, although the deviations from the 45-degree line are considerably smaller. In contrast, the PCA-based models no longer display this behavior for the 48-, 72-, and 96-month windows. It is important to note that these deviations from perfect calibration are still within the confidence bands and therefore not significant, with the exception of the four aforementioned models.

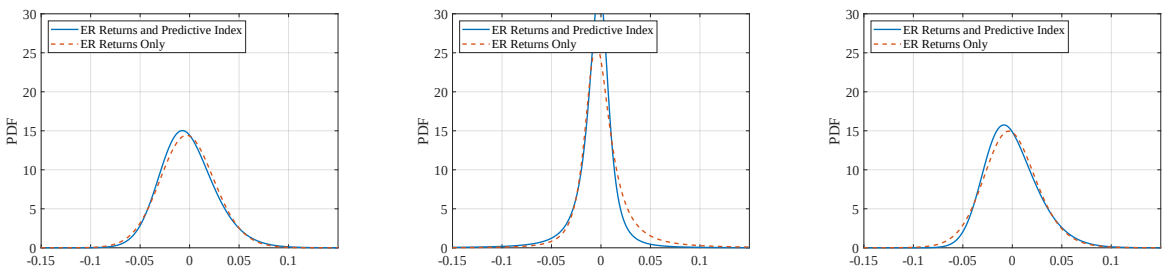
Table 3 reports the test statistics and p-values of the Amisano & Giacomini (2007) test. The test compares the out-of-sample predictive scores of the predicted distribution conditional on a constant, the current log return of the nominal exchange rate and the current value of the predictive index. We use different rolling window sizes to perform out-of-sample forecasting. Furthermore, we use different predictive indices, as specified in Table 4. The test is applied to the LPDS resulting from out-of-sample estimations of these different models. The null of an average difference of zero in predictive scores between the model resulting from the two-step quantile regression approach and the benchmark model can be rejected in both directions. A negative test statistic indicates that the predictive scores of the specified model are higher than that of the benchmark model. Grey brackets are used to highlight insignificant negative test statistics. If the predictive scores of the specified model are significantly higher than that of the random walk model at the 5% level, the bracket is highlighted in green. The green brackets thus present significant negative test statistics. Conversely, positive test statistics indicate that the benchmark model's predictive scores are higher than that of the specified model.

Model	**R = 24**	**R = 48**	**R = 72**	**R = 96**
PCA-based Predictive Indices				
FFI_t	1.5651 (0.1176)	0.8548 (0.3926)	0.5703 (0.5685)	-3.2358 (0.0012)
EFI_t	1.3652 (0.1722)	0.9312 (0.3517)	0.9243 (0.3553)	-0.9366 (0.3490)
CFI_t	1.7727 (0.0763)	0.8972 (0.3696)	-0.8293 (0.4100)	-3.2597 (0.0011)
Relative Yield Predictive Indices				
$LFAAA_t$	2.1794 (0.0293)	1.3434 (0.1791)	0.4789 (0.6320)	-2.8482 (0.0044)
$LFALL_t$	1.7552 (0.0792)	1.5231 (0.1277)	-3.3995 (0.0007)	-2.5778 (0.0099)
$SFAAA_t$	NaN (NaN)	1.2352 (0.2168)	-1.0315 (0.3023)	-2.3963 (0.0166)
$SFALL_t$	2.6131 (0.0090)	1.3397 (0.1803)	-1.9098 (0.0562)	-2.3823 (0.0172)

Table 3: Amisano & Giacomini (2007) test applied to the average LPDS (rolling window sizes specified on top and predictive index specified on the left)

We observe that the random walk density forecasting approach is significantly outperformed by multiple models. At the 2-year and 4-year rolling window size, no model is able to outperform the benchmark model. This suggests that our approach yields inferior density forecasts when

only a limited amount of past information is taken into account. At the 6-year and 8-year rolling window size, most of our introduced models manage to provide a superior set of out-of-sample density forecasts relative to the benchmark model. More specifically, at the 6-year rolling window size, the two-step quantile forecasting approach using the level of the yield curve as predictive index ($LFALL_t$) manages to significantly outperform the benchmark model, resulting in the lowest overall test statistic. Besides, using CFI_t, $SFAAA_t$ and $SFALL_t$ as predictive indices leads to the benchmark model being insignificantly outperformed at R = 72 (as seen in the gray brackets). An interesting result is that although using the Composite Index (CFI_t) leads to an improvement over the benchmark model at the 6-year rolling window size, this is not the case for the Fundamental Index (FFI_t) at the same window. This suggests that the addition of the experimental variables to the Fundamental Index (resulting in the Composite Index) may have added relevant information. At the 8-year rolling window size, the introduced models convincingly beat the benchmark model. Applying the quantile forecasting approach using each predictive index, except for the Experimental Index (EFI_t), provides significantly higher predictive scores than the benchmark model does. As clearly seen in the 8-year window, but also at the 6-year window, using the fundamental index leads to a higher out-of-sample density forecasting accuracy than the experimental index, and is thus a preferred choice for an economic agent.



(a) Pre-crisis (Oct 2006)

(b) Global Financial Crisis (Oct 2008)

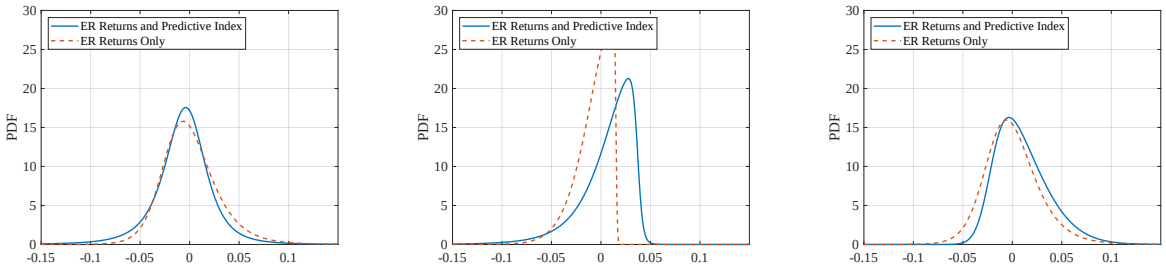
(c) Post-crisis (Oct 2014)

Figure 5: Forecasted one-month-ahead conditional densities of USD/EUR exchange rate returns, shown for the periods before, during, and after the 2008 financial crisis.

Note: One-month-ahead distribution of USD/EUR exchange rate returns conditional on a constant, the current log return of the nominal exchange rate, with and without the current value of the Composite Forecasting Index (CFI) for a 96-month rolling window.

Figure 5 illustrates the estimated conditional density forecasts using a 96-month rolling window at three distinct points in time: pre-crisis (2006M10), during the global financial crisis (2008M10), and post-crisis (2014M10). The comparison highlights notable changes in the shape of the forecasted distributions across periods of varying economic stress. In the pre- and post-crisis periods, the predictive densities, both for the model incorporating the predictive index and the one relying solely on exchange rate returns, exhibit relatively symmetric, bell-shaped curves, resembling a normal distribution. This suggests a stable macroeconomic environment where forecast uncertainty is low and relatively balanced in the upper and lower tails. In contrast, during the crisis period, the density forecast becomes considerably more peaked with high

kurtosis and visible skewness. Notably, the forecast that includes the predictive index deviates more significantly from symmetry, exhibiting a more substantial left tail. This reflects increased dollar depreciation risk as well as the additional sensitivity captured by the predictive index.



(a) Pre-crisis (Oct 2006) (b) Global Financial Crisis (Oct 2008) (c) Post-crisis (Oct 2014)

Figure 6: Forecasted one-month-ahead conditional densities of USD/EUR exchange rate returns using the Eurozone yield level factor, shown for the periods before, during, and after the 2008 financial crisis.

Note: One-month-ahead distribution of USD/EUR exchange rate returns conditional on a constant, the current log return of the nominal exchange rate, with and without the current value of the relative level factor between all Eurozone and US bonds (LF ALL) for a 96-month rolling window.

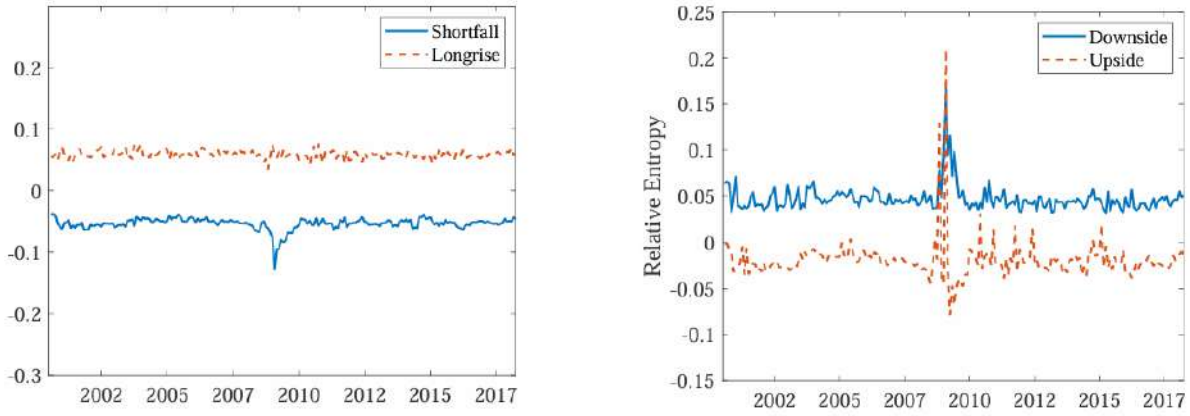
Figure 6 illustrates the forecasted one-month-ahead conditional densities of USD/EUR exchange rate returns across three periods using the relative yield level factor. In the pre-crisis period (Oct 2006), the density appears close to normal, symmetric and no extreme kurtosis indicating a relatively stable forecast environment. During the global financial crisis (Oct 2008), the shape of the predictive density undergoes a significant transformation: it becomes sharply peaked and irregular. The forecast combining exchange rate returns and the relative yield level factor diverges significantly from the returns-only model. This suggests that the inclusion of bond market information during the crisis introduces more concentrated and asymmetric risk expectations. This asymmetric shape can be interpreted as the model incorporating investor expectations embedded in the relative yield level factor, suggesting an increased probability of dollar appreciation. The resulting predictive density reflects this directional signal, concentrating more mass on positive returns for the USD/EUR rate.

In the post-crisis period (Oct 2014), while the density becomes less peeked, it displays noteable skewness and a fatter right tail. This suggests that in the Post-crisis period, the relative yield level factor model estimates higher upside risk and an appreciation of the dollar against the Euro. This contrasts with the CFI-based forecasts, where post-crisis densities return to near pre-crisis, symmetric densities.

5.2 Risk Metrics: Expected Shortfall, Expected Longrise, and Entropy

In Figure 7, we present the evolution of the upside and downside entropy for the exchange rate returns. Besides, we plot the evolution of the 5% Expected Shortfall and the 95% Expected Longrise. Given that computing these risk measures is computationally expensive, we restrict our analysis to the Composite Forecasting Index (CFI_t) using the 96-month rolling window. The

window is chosen because it yielded the most pronounced results in out-of-sample forecasting across our set of predictive indices. The specific index is selected because the composite index achieves the lowest out-of-sample test statistic at the 96-month window, as reported in Table 3.



(a) Expected Shortfall

(b) Entropy

Figure 7: Time series evolution of the 5% expected shortfall ES_τ and the relative downside and upside entropy $\mathcal{L}_t^D$ and $\mathcal{L}_t^U$

The upside entropy $\mathcal{L}_t^U$ and downside entropy $\mathcal{L}_t^D$ exhibit relatively strong co-movements, even during the global financial crisis of 2008-2009. However, the level of the downside entropy is almost consistently higher than that of the upside entropy. This indicates that the conditional distribution continually assigns more probability mass to extreme deprecations of the dollar relative to the euro than the unconditional distribution does. However, this is not the case for extreme appreciations of the dollar relative to the euro. The graph thus implies that, given the current macroeconomic and financial conditions stated by the Composite Index, large depreciations of the dollar are more likely to occur than suggested by the unconditional distribution.

Additionally, the Expected Longrise appears to remain relatively stable throughout the sample period, showing no notable upward or downward trend and only minor short-term variability. This suggests that the extreme upward risk, i.e., the expected appreciation in the upper 5% quantile of the dollar relative to the euro, remains relatively stable over time. Similarly, the Expected Shortfall remains fairly constant over time. There is, however, a notable dip around 2008, corresponding to the global financial crisis. This indicates an elevated downside risk for the dollar i.e., a stronger depreciation of the dollar relative to the euro in the lowest 5% percentile of the distribution. However, this spike does not persist as the Expected Shortfall returns to its previous level after the crisis.

6 Conclusion and Discussion

This paper aims to examine whether one-month-ahead conditional density forecasts for the USD/EUR exchange rate returns, constructed using a quantile interpolation framework, can outperform those of a random walk. To improve upon an unconditional density forecast, seven different index specifications are proposed as conditional variables for future exchange rates.

Three of the indices are constructed using macro-economic and financial data, whilst four of the indices are based on the relative yield curves of the Eurozone and the US. Following an Amisano & Giacomini (2007) - test, it can be concluded that almost all constructed conditional density forecasts significantly outperform those of a random walk model at a 96-month rolling window. Density forecasts conditional on the level of a relative yield curve outperform those of a random walk model at the 72-month rolling window. All results are statistically assessed for proper calibration using a PITs test.

Our paper argues that the entire distribution of exchange rate returns evolves over time. Conditioning on a macro-financial index for the US and Eurozone leads to the distribution continually assigning a higher probability mass to extreme depreciations of the dollar relative to the euro over time. A univariate quantile regression analysis reveals how an increase in the yield spread between the Eurozone and the US is associated with an appreciation of the dollar relative to the euro. Overall, this paper offers new perspectives into the longstanding debate on possible determinants of future exchange rates. Furthermore, our results show that a straightforward density forecasting approach without underlying parametric assumptions performs well out-of-sample, providing a new insight into the 'Meese-Rogoff puzzle'.

Our paper contains several limitations. Some of the macroeconomic indicators used to create conditional variables suffer from substantial publication lags, reducing their real-time relevance for forecasting. Second, the usage of Principal Component Analysis to construct a predictive index, whilst effective in reducing dimensionality, limits interpretability of the importance of individual variables. Besides, it is performed on the entire available sample of macro-financial variables, which would be impossible when constructing a forecast beforehand. Future research could address these issues in several ways. Nowcasting techniques can help address reporting lags by updating forecasts as data becomes available. Alternative index summarization techniques, such as dynamic factor models or time-varying parameter models such as the TVP-FAVAR model by Koop & Korobilis (2014), could prevent reliance on future data. Additionally, extending the framework to multi-horizon forecasting could be valuable for forward-looking variables like yield curve factors, potentially revealing varying predictive power across different forecast horizons.

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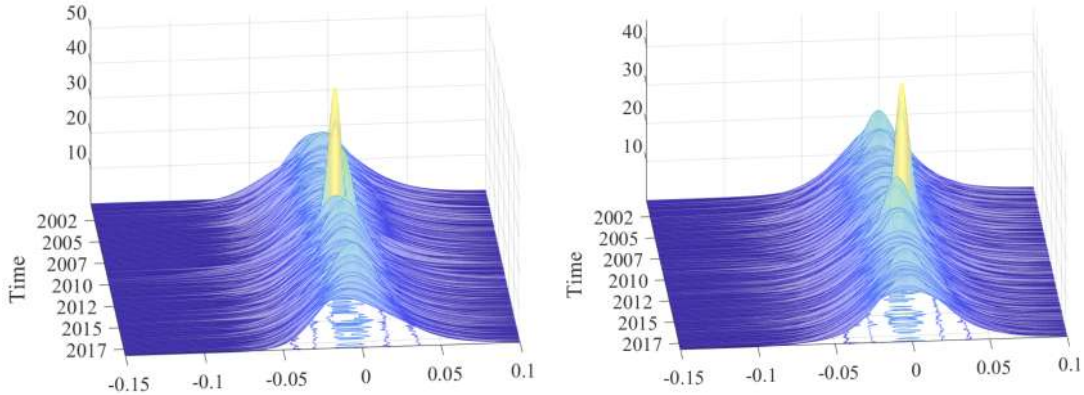
A Programming code

See replication package. Our code builds upon the replication package provided by Adrian et al. (2019a), which we adapted to suit our forecasting framework.

B Overview transformed variables: Table 4

C Other Figures In-Sample Analysis

C.1 Conditional Distribution FFI and EFI

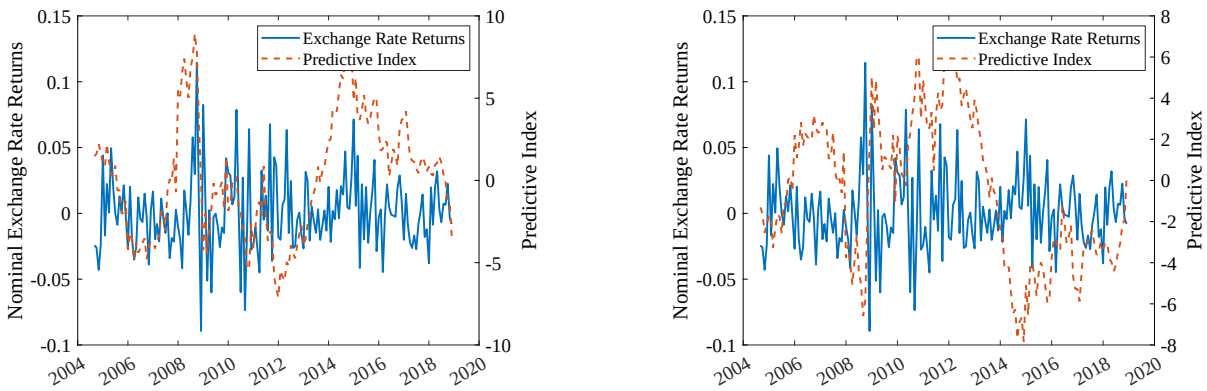


(a) FFI (b) EFI

Figure 8: In-sample conditional distribution of USD/EUR exchange rate log return over time

Note: One-month-ahead model fit distribution of USD/EUR exchange rate returns conditional on a constant, the current log return of the nominal exchange rate and the noted predictive index.

C.2 Forecasting Indices



(a) LF AAA (b) SF AAA

Figure 9: Time series of USD/EUR exchange rate log returns and predictive indices (LF AAA and SF AAA)

Table 4: Overview transformed data

Variable	Formula Transformation
Transformed data for (1) Fundamental and (3) Composite forecasting index	
CPI_EZ_logdiff$_t$	log(EZ Consumer Price Index for month t) $-$ log(EZ Consumer Price Index for month $t-1$)
CPI_logdiff_USA_minus_EZ$_t$	[log(USA CPI for month t) $-$ log(USA CPI for month $t-1$)]$-$ CPI_EZ_logdiff$_t$
interest10Y_minus_interbank3m_USA$_t$	10-year yield USA - 3-month interbank rate USA, both at latest available day of month t
interest10Y_minus_interbank3m_EZ$_t$	10-year yield Eurozone - 3-month interbank rate Eurozone, both at latest available day of month t
NFCI$_t$	NFCI of latest full week in month t
IndustrialProd_USA_logdiff$_t$	log(USA Industrial Production in month t) $-$ log(USA Industrial Production in month $t-1$)
IndustrialProd_EZ_logdiff$_t$	log(EZ Industrial Production in month t) $-$ log(EZ Industrial Production in month $t-1$)
GDP_USA_logdiff$_t$	log(USA GDP last quarter before t) $-$ log(USA GDP last quarter before $t-1$)
GDP_EZ_logdiff$_t$	log(EZ GDP last quarter before t) $-$ log(EZ GDP last quarter before $t-1$)
DebtToGDP_USA_logdiff$_t$	log(USA Debt-to-GDP last quarter before t) $-$ log(EZ GDP last quarter before $t-1$)
DebtToGDP_EZ_logdiff$_t$	log(EZ Debt-to-GDP last quarter before t) $-$ log(EZ GDP last quarter before $t-1$)
NetCA_USA_minus_EZ$_t$	USA Total Net Current Account last quarter before $t-$ EZ Total Net Current Account last quarter before t
Transformed data for (2) Experimental Variables and (3) Composite forecasting index	
SP500_logdiff$_t$	log(S&P500 last trading day of month t) $-$ log(S&P500 last trading day of month $t-1$)
EUROSTOXX50_logdiff$_t$	log(EUROSTOXX50 last trading day of month t) $-$ log(EUROSTOXX50 last trading day of month $t-1$)
OilPrice_WTI_logdiff$_t$	log(WTI Crude Oil Index at end of month t) $-$ log(WTI Crude Oil Index at end of month $t-1$)
Selected data for (4, 5, 6, 7) Relative Yield Forecasting indices	
Yields_UST	Market Yield on U.S. Treasury Securities at Constant Maturity, Quoted on an Investment Basis (DGS): 3-Month, 6-Month, 1-Year, 2-Year, 3-Year, 5-Year, 7-Year, 10-Year, 20-Year, and 30-Year
Yields_EZ_AAA	Yield curve data, "AAA-rated" euro area central government bonds: spot rate for 3-month, 6-month, 1-Year, 2-Year, 3-Year, 5-Year, 7-Year, 10-Year, 20-Year, and 30-Year
Yields_EZ_ALL	Yield curve data, all (including AAA-rated) euro area central government bonds: spot rate for 3-month, 6-month, 1-Year, 2-Year, 3-Year, 5-Year, 7-Year, 10-Year, 20-Year, and 30-Year

Note: 'last quarter before t' means the most recently finished quarter in month t,
i.e. Q1 for April, May, and June; Q2 for July, August, September; Q3 for October, November, December; and Q1 for January, February, March

C.3 Univariate Quantile Regressions

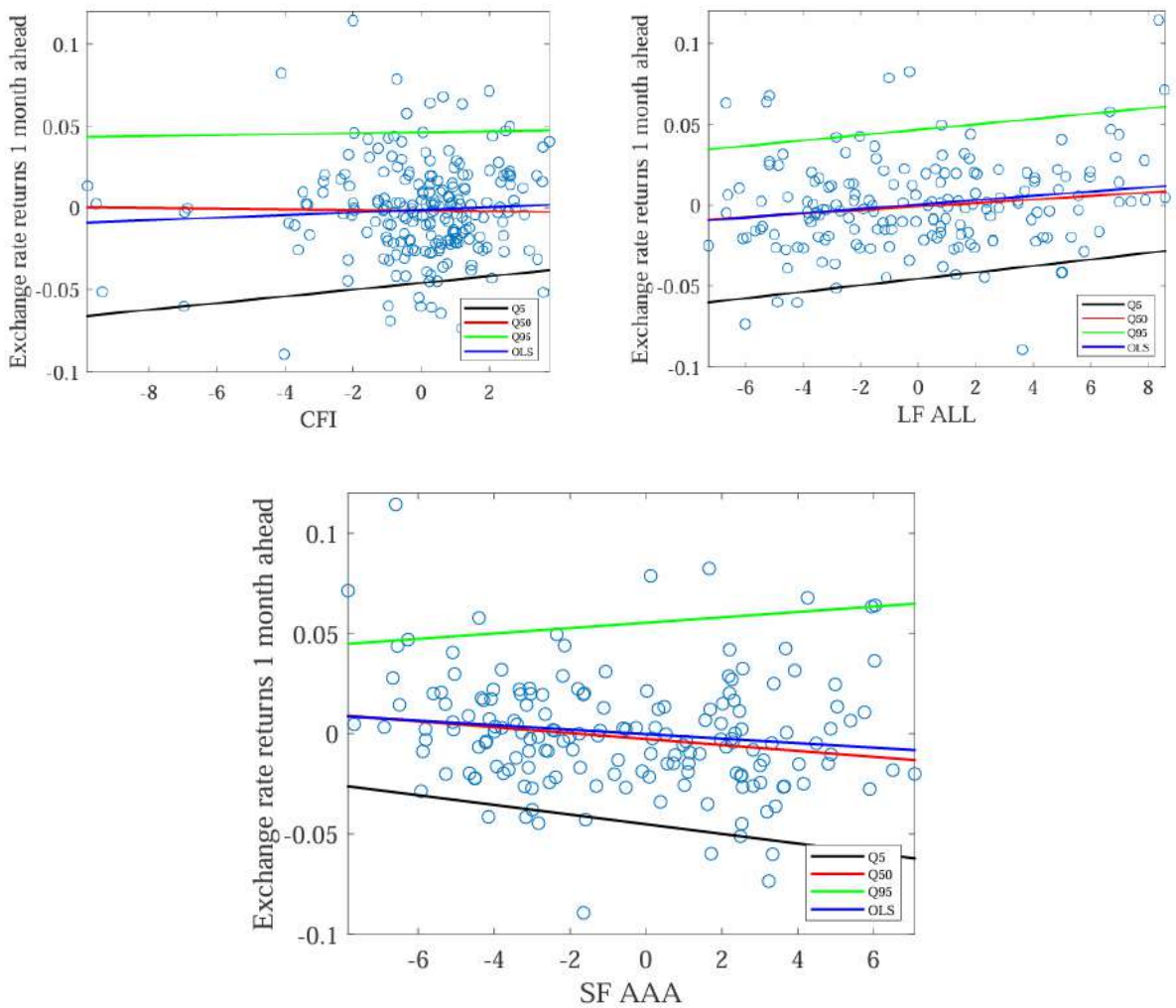
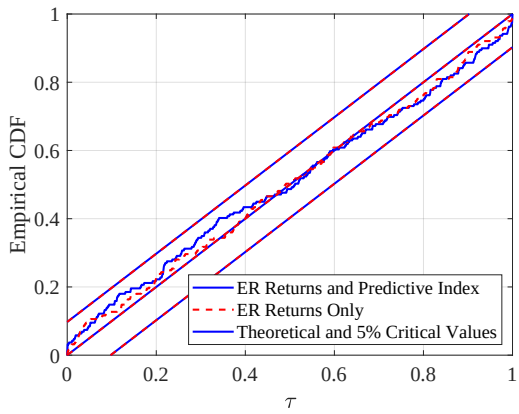


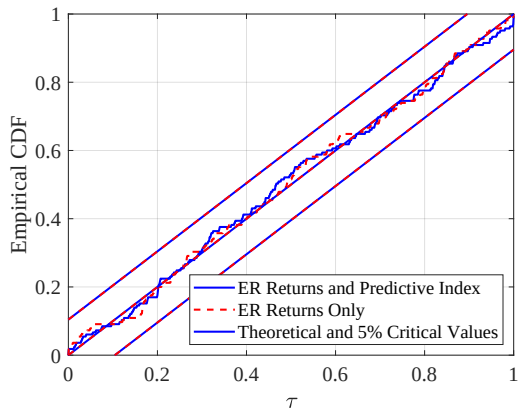
Figure 10: In-sample univariate quantile regressions of the one-month-ahead log exchange rate returns on the composite index, the slope factor of the Eurozone AAA-rated - US Treasury bonds $SFAAA_t$ and the level factor of the Eurozone (all) - US Treasury bonds $LFALL_t$

D Out-of-Sample Calibration Figures

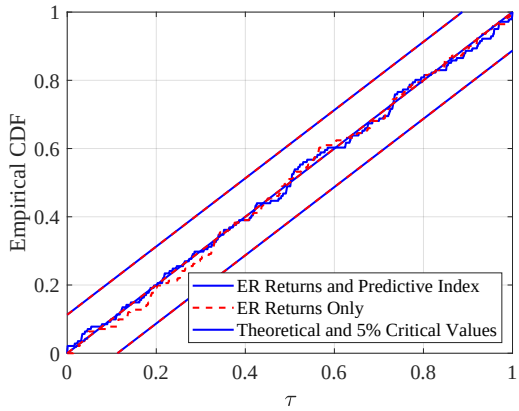
D.1 Fundamental Forecasting Index (FFI)



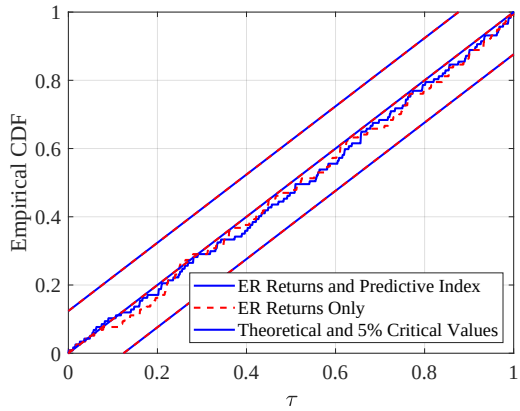
(a) 24-month rolling window



(b) 48-month rolling window



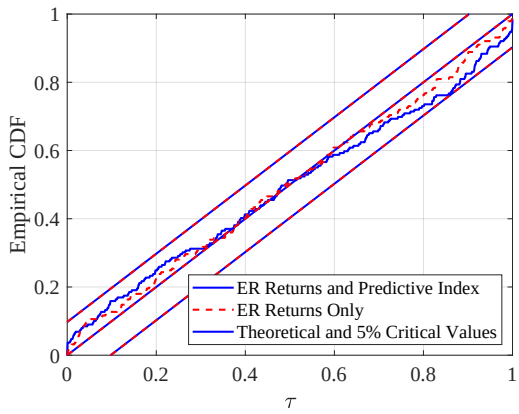
(c) 72-month rolling window



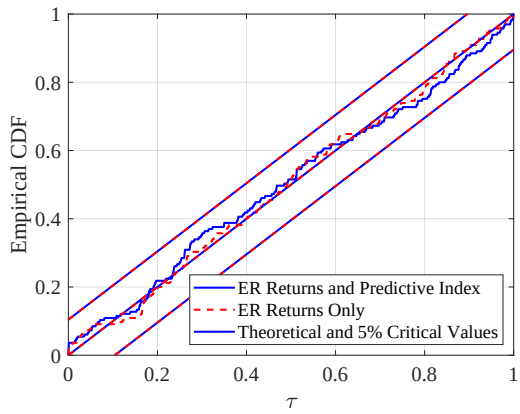
(d) 96-month rolling window

Figure 11: PIT Calibration Plots — Fundamental Forecasting Index (FFI)

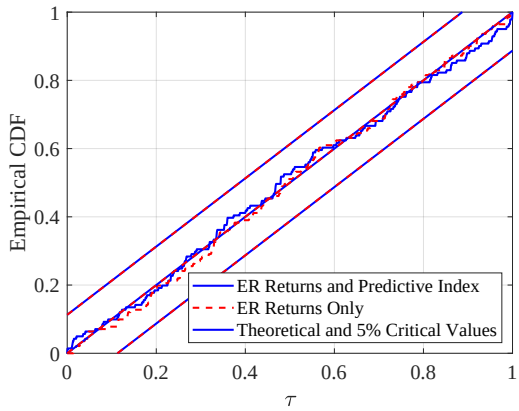
D.2 Experimental Forecasting Index (EFI)



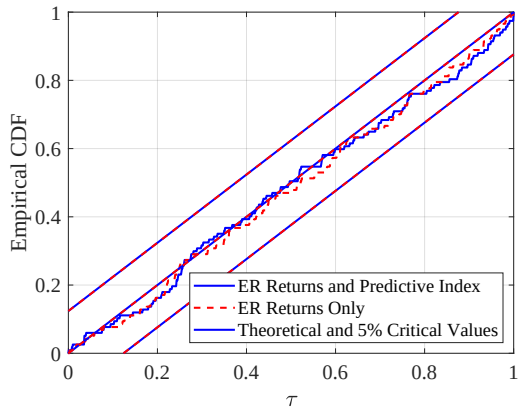
(a) 24-month rolling window



(b) 48-month rolling window



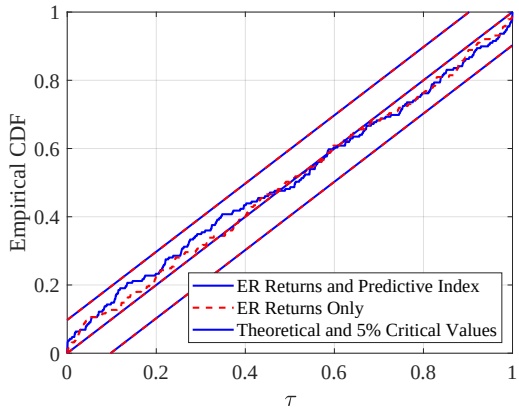
(c) 72-month rolling window



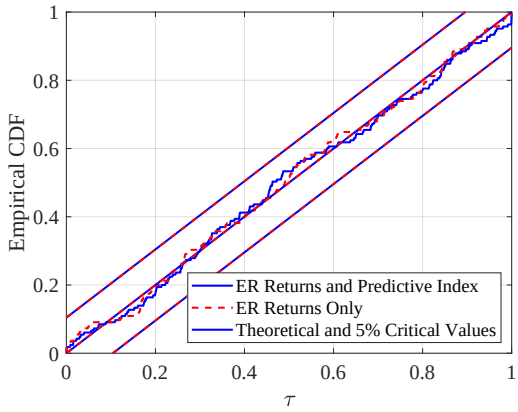
(d) 96-month rolling window

Figure 12: PIT Calibration Plots — Experimental Forecasting Index (EFI)

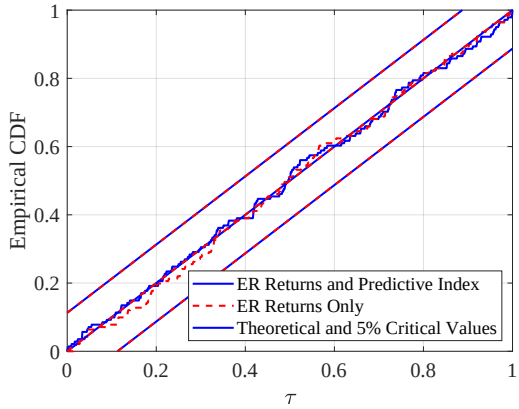
D.3 Composite Forecasting Index (CFI)



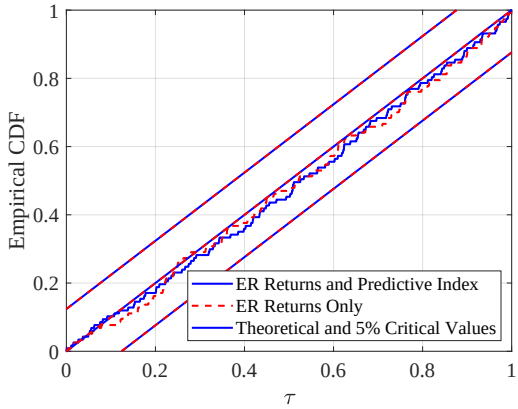
(a) 24-month rolling window



(b) 48-month rolling window



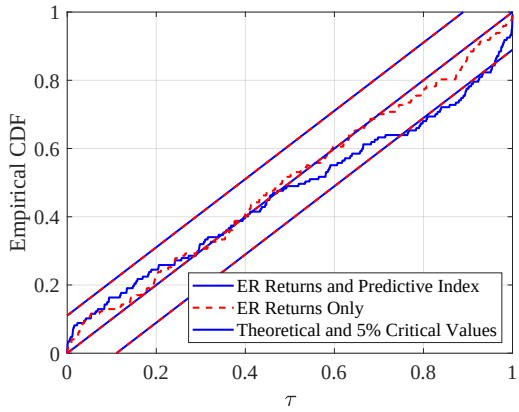
(c) 72-month rolling window



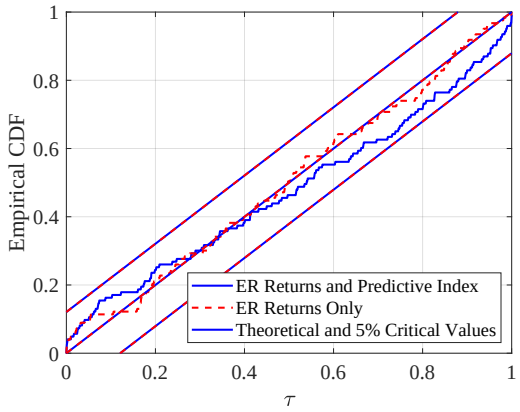
(d) 96-month rolling window

Figure 13: PIT Calibration Plots — Composite Forecasting Index (CFI)

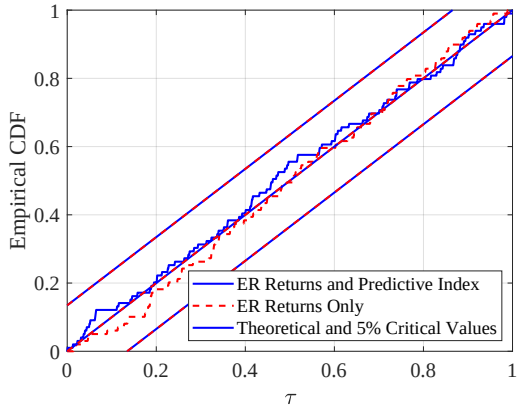
D.4 Relative Level Factor (AAA-rated Bonds)



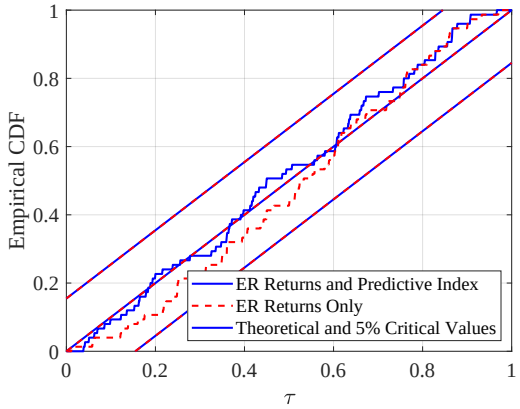
(a) 24-month rolling window



(b) 48-month rolling window



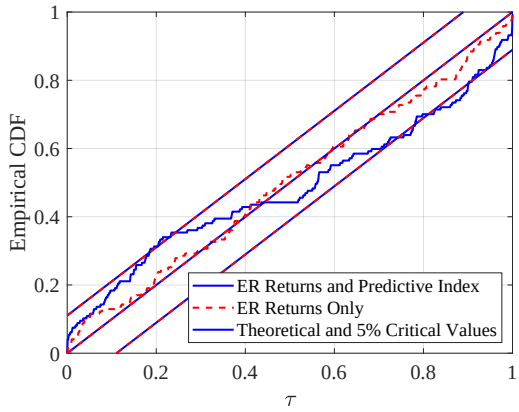
(c) 72-month rolling window



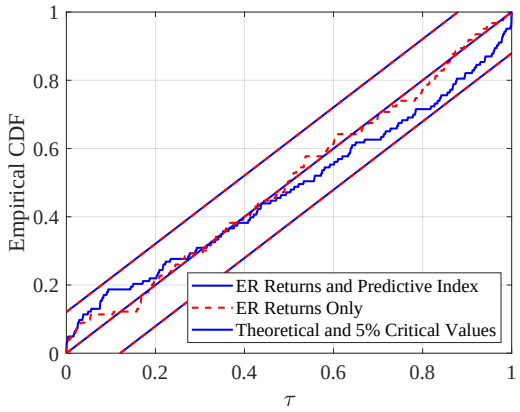
(d) 96-month rolling window

Figure 14: PIT Calibration Plots — Relative Yield Level Factor (AAA-rated Eurozone - US)

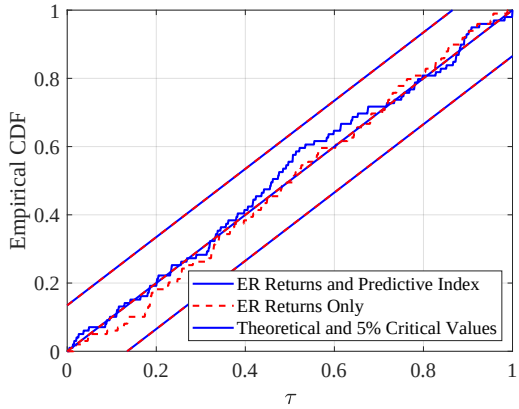
D.5 Relative Level Factor (All Eurozone Bonds)



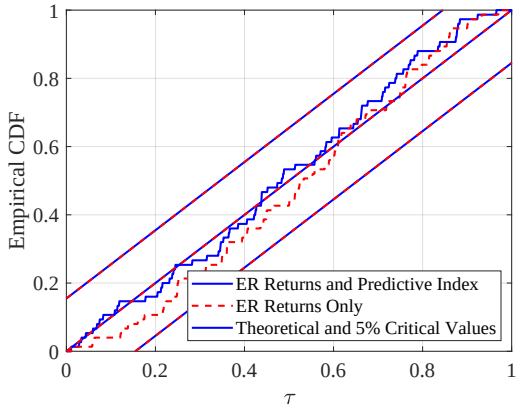
(a) 24-month rolling window



(b) 48-month rolling window



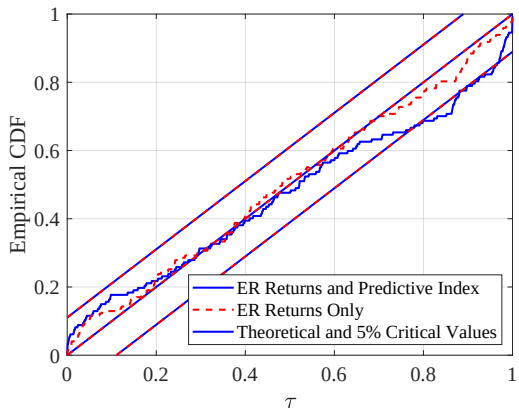
(c) 72-month rolling window



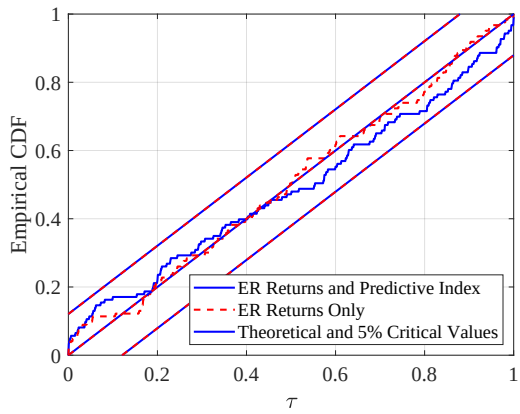
(d) 96-month rolling window

Figure 15: PIT Calibration Plots — Relative Yield Level Factor (All Eurozone - US)

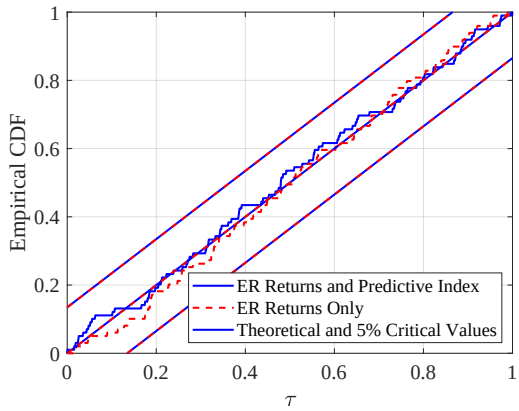
D.6 Slope Factor (AAA-rated Bonds)



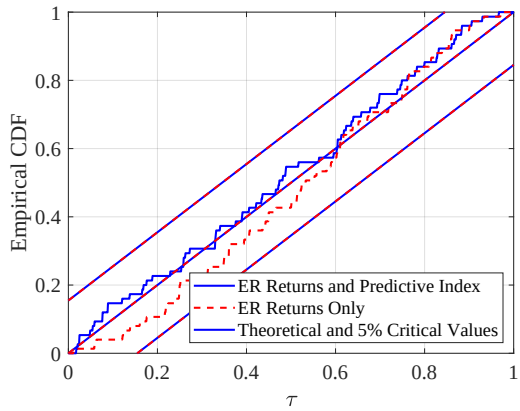
(a) 24-month rolling window



(b) 48-month rolling window



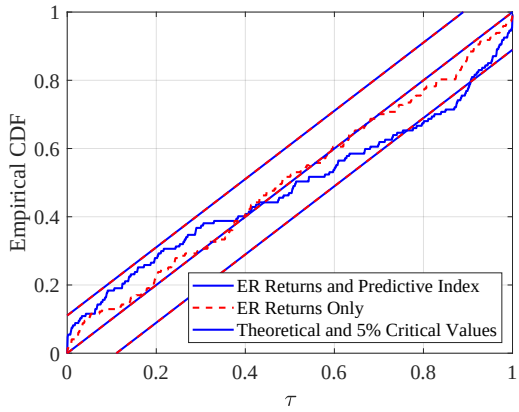
(c) 72-month rolling window



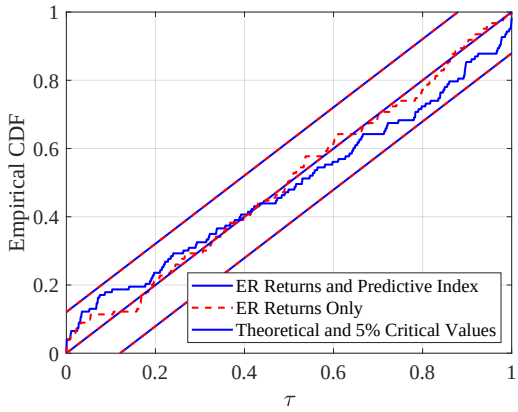
(d) 96-month rolling window

Figure 16: PIT Calibration Plots — Slope Yield Curve Factor (AAA-rated Eurozone - US)

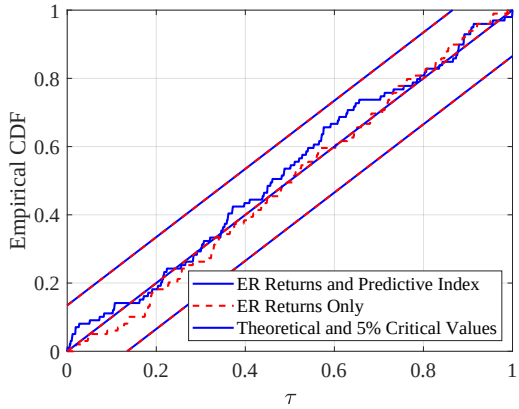
D.7 Slope Factor (All Eurozone Bonds)



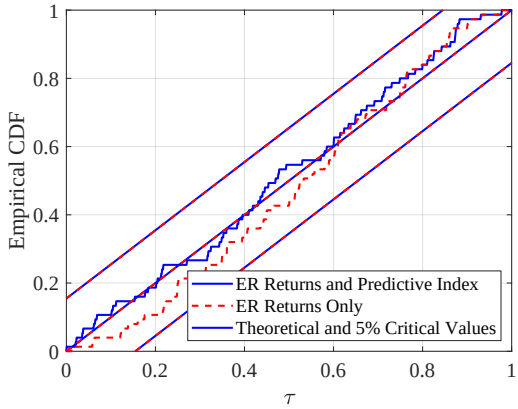
(a) 24-month rolling window



(b) 48-month rolling window



(c) 72-month rolling window



(d) 96-month rolling window

Figure 17: PIT Calibration Plots — Slope Yield Curve Factor (All Eurozone - US)